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Cellular adaptation to cancer therapy along a resistance continuum

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Advancements in precision oncology over the past decades have led to new therapeutic interventions, but the efficacy of such treatments is generally limited by an adaptive process that fosters drug resistance[1]. In addition to genetic mutations[2], recent research has identified a role for non-genetic plasticity in transient drug tolerance[3] and the acquisition of stable resistance[4,5]. However, the dynamics of cell-state transitions that occur in the adaptation to cancer therapies remain unknown and require a systems-level longitudinal framework. Here we demonstrate that resistance develops through trajectories of cell-state transitions accompanied by a progressive increase in cell fitness, which we denote as the 'resistance continuum'. This cellular adaptation involves a stepwise assembly of gene expression programmes and epigenetically reinforced cell states underpinned by phenotypic plasticity, adaptation to stress and metabolic reprogramming. Our results support the notion that epithelial-to-mesenchymal transition or stemness programmes—often considered a proxy for phenotypic plasticity—enable adaptation, rather than a full resistance mechanism. Through systematic genetic perturbations, we identify the acquisition of metabolic dependencies, exposing vulnerabilities that can potentially be exploited therapeutically. The concept of the resistance continuum highlights the dynamic nature of cellular adaptation and calls for complementary therapies directed at the mechanisms underlying adaptive cell-state transitions.

The ability of malignant cells to consistently evolve resistance to initially effective therapies poses one of the major obstacles to cancer treatment[1]. Models to explain the emergence of resistance have traditionally relied on the selection of genetic mutations enabling cells to escape therapy[2]. More recently, however, the roles of non-genetic mechanisms have been increasingly recognized[5]. In particular, a subset of cancer cells can avoid therapeutic elimination by entering a drug-tolerant persister cell state through transcriptional reprogramming[3,6,7]. Although such a drug-tolerant state may be initially reversible, over long-term exposure to the drug, heritable adaptive states can eventually impart stable resistance[8–10].

Recent evidence also suggests that drug resistance unfolds as a gradual and multifactorial process[11]. However, the underlying dynamics and the phenotypic trajectories that mediate drug adaptation remain unknown. Here we systematically probe and functionally characterize these trajectories in a series of longitudinal experiments. We demonstrate that drug adaptation from initial sensitivity to stable resistance can be conceptualized as a series of transitions between distinct and progressively adaptive cellular states, which we term the 'resistance continuum'.

Drug adaptation along a resistance continuum

To study the dynamics of the adaptive potential of cancer cells, we reasoned that long-term dose escalation coupled with single-cell and functional tools would provide an experimental framework to reveal the gene programmes associated with the emergence of resistance. We treated a human *BRCA2*-deficient high-grade serous ovarian cancer (HGSOC) cell line (Kuramochi[12]) with the PARP-inhibitor (PARPi) olaparib, which induces synthetic lethality in tumours deficient in *BRCA1* or *BRCA2*[13]. Drug-naive cells were initially treated with 1 μM of olaparib—a relatively low dose—until the population reached confluency, indicating adaptation to the dose. The resulting population was then seeded onto a fresh plate and treated with an increased dose of 2.5 μM.

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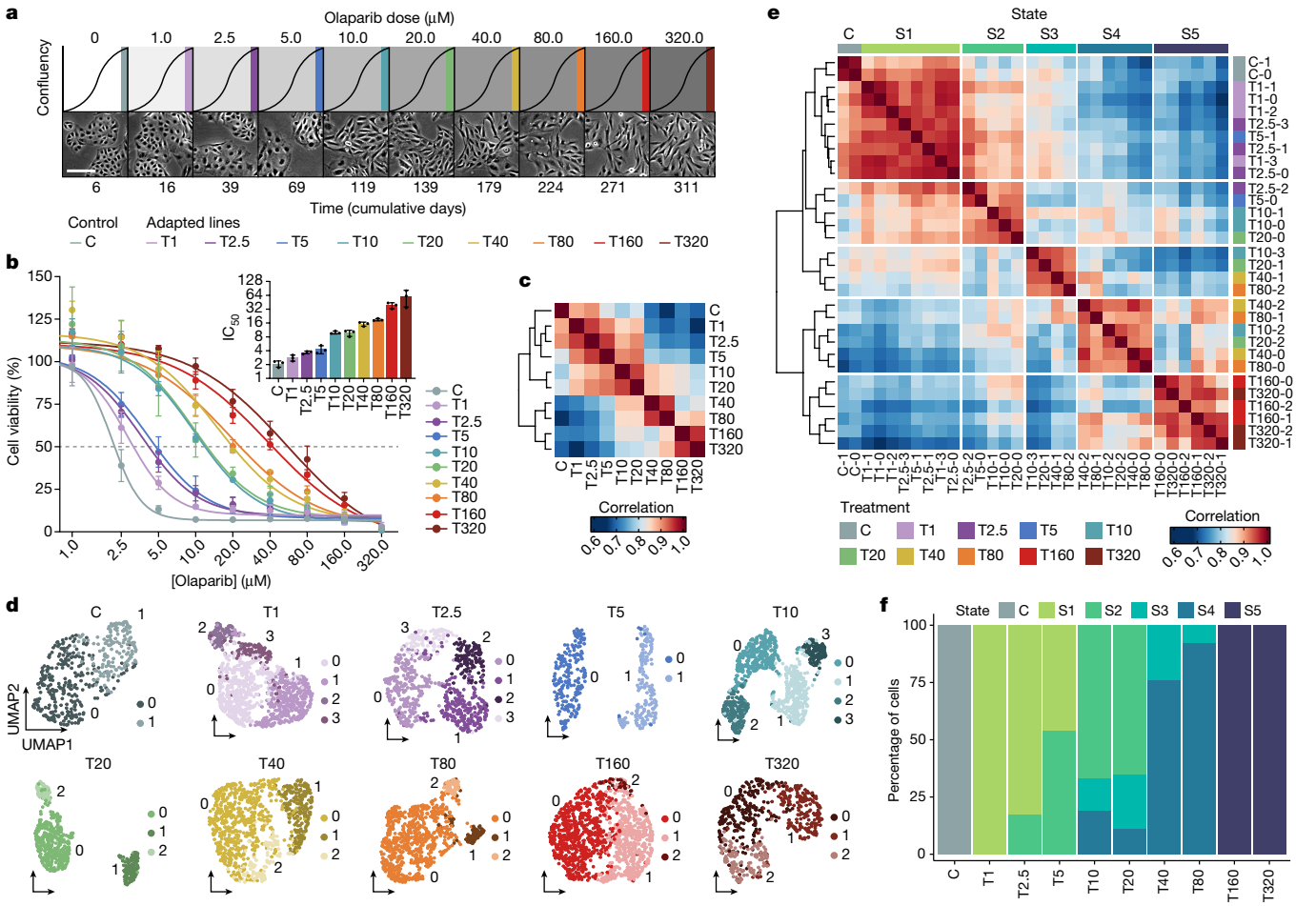


Fig. 1 | Progressive cell-state transitions along a resistance continuum. **a**, Schematic of the experimental design for the generation of the drug-adapted lines. Specific doses and durations of treatments are indicated. Cell morphology is shown from representative microscopic images (scale bar 50 μm, one replicate). **b**, Cell viability of the olaparib response during 9 days of treatment. Data points are normalized relative to vehicle-treated controls for each respective line and represent the average of 3 independent experiments (6 technical replicates per experiment) with their standard deviations. **c**, Spearman's correlations among the averaged transcriptomes of the adapted cell lines. **d**, Uniform manifold approximation and projection (UMAP) representation of scRNA-seq data for the individual lines. Colours and numbers indicate subpopulations determined by Louvain clustering. **e**, Clustering of subpopulations based on their Spearman's rank correlation coefficients across the adaptive lines. Five major transcriptional states are defined (S1–S5). **f**, Frequency of cells in each of the five states across the adaptive lines. Cells were assigned to their respective states on the basis of their subpopulation clustering shown in **e**.

We then repeated this process, each time doubling the drug concentrations, until cells were able to reach confluency at an extreme dose of 320 μM, thereby generating a panel of 9 adapted cell populations (T1 to T320) (Fig. 1a). We found that these populations (hereafter 'lines') became progressively more resistant, with the half maximal inhibitory concentration (IC_{50}) increasing from around 2 μM for the control population to around 60 μM for the cells adapted to 320 μM (Fig. 1b). This shift indicates an increase in drug tolerance toward higher doses, rather than full resistance, as even the most resistant lines continued to exhibit sensitivity to the doses to which they had adapted.

To assess the generality of the observed progressive fitness increases, we repeated the dose-escalation experiment for two additional *BRCA*-mutant ovarian cancer cell lines, Ovsaho and COV362[12], a melanoma cell line (A375) treated with dabrafenib (a BRAF inhibitor), and a lung adenocarcinoma line (PC9) treated with osimertinib (an EGFR inhibitor). We found that the dose-escalation experiments across these distinct models and therapies reproducibly generated continuous resistant phenotypes (Extended Data Fig. 1a–h).

To test for the possibility that pre-existing fully resistant clones were already present in the initial population and selected in the course of our assay, we performed two rounds of treatment-recovery experiments on drug-naive cells under the highest dose exposure (320 μM). A prediction of this scenario is that cells would exhibit increased resistance after the first and/or second rounds. However, we observed that the surviving cells maintained a sensitive state, suggesting a drug-tolerant persister phenotype[3] (Extended Data Fig. 2a). We next tested whether the duration of the treatment alone in the adapted lines could account for increased resistance. We found that prolonged exposure at constant doses contributed to increased resistance but did not elicit similar resistance levels to the dose-escalated populations, especially for the less-adapted lines (Extended Data Fig. 2b–d). We further tested whether dose escalation facilitates adaptation by comparing drug-naive populations exposed to acute doses with dose escalation and found that the dose escalation resulted in faster adaptation (Extended Data Fig. 2e–h), suggesting that lower doses may prime cells to achieve higher resistance.

Overall, these results indicate that gradual drug exposure elicits incremental adaptive responses unfolding along a resistance continuum. We therefore hypothesized that this fitness increase is accompanied by a series of cellular state transitions occurring along an adaptive path.

Drug-induced cell-state transitions

We applied single-cell RNA sequencing (scRNA-seq) to characterize the cellular states along the resistance continuum. Comparing the average

transcriptomes across the ten lines, a similarity dendrogram recovered the treatment history, indicating the divergence of the adapted lines (Fig. 1c and Extended Data Figs. 3a,f and 4a,g). Examining single-cell transcriptomes, we identified distinct subpopulations within the lines (Fig. 1d and Methods) and compared them to inquire whether they comprise shared states. By clustering their average profiles, we found that subpopulations coalesced into distinct states (Fig. 1e). Notably, subpopulations from earlier adaptive lines (such as T2.5-2 and T5-0) clustered with those from later ones (such as T10-0 and T20-0, in state S2), indicating cell-state variability[14] within and across the lines. Along the resistance continuum, multiple states emerge and shifts in frequency reveal differential fitness (Fig. 1f). Repeating this analysis across other resistance models revealed similar dynamics of progressive cell-state transitions (Extended Data Figs. 3b–d,g–i and 4b–e,h–k).

We next sought to identify the gene expression programmes that underpin these adaptive states. Implementing a gene module detection approach, we defined six modules (Fig. 2a). The modules were enriched with genes of distinct functional annotations, many of which have been previously identified in cell survival programmes in response to therapy[15] (Fig. 2b). Module A, expressed by the less-adapted cells, included the lineage-defining transcription factors *SOX17*, *PAX8* and *WT1*[16] as well as epithelial markers (*KRT8* and *DSP*). The expression of these genes decreases along the resistance continuum, reflecting dedifferentiation (Fig. 2a), which was also observed in the other ovarian cancer models (Extended Data Fig. 3e,j). Module A is also enriched for the interferon response (*IFI6* and *IFI27*), an endogenous signature that is induced upon DNA damage[17] (Fig. 2a,b). Module B genes include epithelial (*KRT8* and *KRT18*) and mesenchymal markers (*FN1*), suggesting a hybrid state. Module C includes genes involved in extracellular matrix remodelling (*COL3A1* and *SERPINE2*), epithelial-to-mesenchymal transition (EMT) (*SOX4* and *VIM*) and hypoxia (Fig. 2a,b). Module D also includes hypoxia genes (*CITED2*, *SGK1* and *DUSP1*), EMT (*SNAI2* and *VIM*), mediators of TGFβ signalling (*SMAD3* and *DAB2*) and stemness markers (*CD44* and *KLF4*), suggesting a spectrum of dedifferentiated states (Fig. 2a,b and Extended Data Fig. 3e,j). Indeed, we observed morphological changes to mesenchymal-like phenotypes early on in the adaptive process (Fig. 1a) and found high expression of those markers in the resistant populations from the additional ovarian cancer models (Extended Data Fig. 3e,j and Supplementary Tables 2 and 3). Studying other resistance models confirmed dedifferentiation as a general feature of drug adaptation (Extended Data Figs. 3e,j and 4f,l).

The genes assigned to modules E and F reflect metabolic rewiring in the highly resistant cells in state S5. These include the hypoxia regulator *HIF1A* and its target *DDIT4*, as well as glycolytic enzymes (*PFKP* and *GAPDH*). Notably, we observed enrichment of *NRF2* cytoprotective programmes for oxidative stress (*TXNRD1*, *AKR1B1* and *GGH*) and ferroptosis (*HMOX1*, *GPX4* and *FTL*), which are associated with survival advantage in drug-tolerant persisters[7]. NRF2 regulates the pentose phosphate pathway (PPP) (*TKT*), which contributes to maintaining redox balance and provides precursors for de novo nucleotide biosynthesis (*IMPDH2*) (Fig. 2a,b). A number of ribosomal (*RPL14* and *RPS15A*) and mRNA translation genes (*EIF1*, *EIF2D* and *RACK1*) in modules D and E suggests translation reprogramming (Supplementary Table 1) controlled by ATF4, a transcription factor that cooperates with NRF2 in the integrated stress response[18]. Genes regulated by NRF2 and ATF4 (*ASNS*, *CHAC1* and *SHMT2*) were consistently detected in the highly resistant populations from the additional ovarian, lung and melanoma models (Extended Data Figs. 3e,j and 4f,l and Supplementary Tables 2–5).

As the resistance continuum implies that state transitions are causally related to fitness, we tested whether, within and across adapted lines, cells have differential fitnesses according to their states. As our gene module definitions identified CD24 and CD44 as markers for specific states (Extended Data Fig. 5a), we sorted subpopulations from a representative set of lines (C, T10, T40 and T320; Extended Data Fig. 5b). Indeed, we found that within the T10 population, state

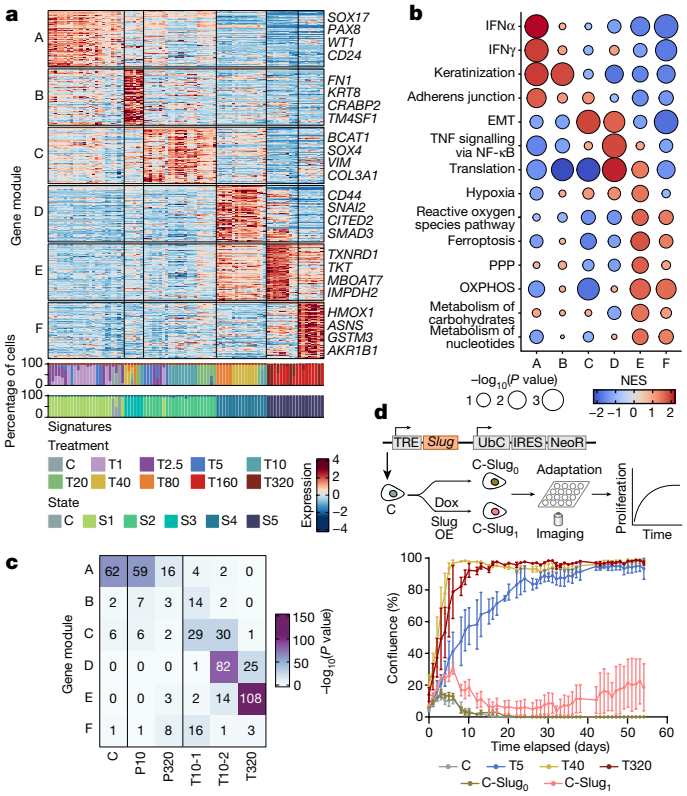


Fig. 2 | Assembly of gene module expression accompanies drug adaptation. **a**, Heat map indicating gene modules associated with distinct transcriptional states: genes (rows) and cell signatures (columns). Bar plots (bottom) show the frequency of cells from each signature and state (see Supplementary Fig. 1a,b and Supplementary Table 1). **b**, Gene set enrichment analysis (GSEA) of each gene module ($P < 0.01$ and false discovery rate (FDR) < 0.25 for multiple test correction, one-sided Kolmogorov–Smirnov test). The colour scale is the normalized enrichment score (NES) as calculated by GSEA and circle sizes indicate *P* values. OXPHOS, oxidative phosphorylation. **c**, Gene overlap between the differentially expressed genes across persister and adapted lines with the gene modules. Colour scale indicates *P* values from one-sided hypergeometric tests. **d**, Schematic of the inducible *Slug* construct (top). Quantification by live-cell imaging of the percentage of area covered by cells treated with 5 μM olaparib over a 54-day time period (bottom). Growth curves compare the parental C, adapted T5, T40 and T320 and *Slug*-induced C-Slug$_0$ (control) and C-Slug$_1$ (1 μg ml^{-1} doxycycline). Data are mean ± s.d. of 3 replicates shown (3 technical replicates each). Growth curve under DMSO is shown in Extended Data Fig. 6m.

S4 cells (CD24^{low}CD44^{hi}) were more resistant than the state S2 cells (CD24^{low}CD44^{low}; Extended Data Fig. 5c), but were less resistant than the CD24^{low}CD44^{hi} cells from T40 and T320. This indicates that suboptimal cellular states coexist alongside states of higher fitness within a population, whereas fitness increases across the adapted populations.

We next tested whether transcriptional changes in the adapted lines could be explained exclusively by exposure to acute high doses. We generated populations of drug-tolerant persisters that survived treatment with 10 μM of olaparib (hereafeter designated P10; sufficient to decrease viability by 90% in drug-naive cells; Fig. 1b) and 320 μM (hereafter, P320) (see Supplementary Fig. 2 and Supplementary Discussion). We found that acute doses induced persister-specific states (Extended Data Fig. 3k), an observation also noted when comparing PC9 persister and adapted lines (Extended Data Fig. 4i,j). The set of differentially expressed genes in P10 and P320 (relative to the parental control cells) strongly overlapped with gene module A (expressed by untreated and state S1 cells) (Fig. 2c). Genes associated with adapted states such as *CD44*, *HMOX1*, *GPX4*, *DDIT4* and *PFKP* were highly expressed

in persisters, albeit at lower levels compared with the adapted lines (Extended Data Fig. 3l and Supplementary Table 6), suggesting partial reprogramming that may prime subsequent adaptive paths.

Lineage plasticity enables drug adaptation

Our transcriptome data identified dedifferentiation as one axis within the resistance continuum, as we observed: (1) loss of expression of the lineage factor *SOX17* (Fig. 2a); (2) upregulation of EMT programmes (Fig. 2a,b); and (3) a $CD24^{low}CD44^{hi}$ phenotype (Extended Data Fig. 5), a feature associated with enhanced plasticity and stem-like states[19]. Thus, we aimed to investigate the effect of perturbing these programmes on resistance. We induced *SOX17* expression in the most adapted line (T320) and observed an increase in the sensitivity to olaparib (Extended Data Fig. 6a,b). Conversely, decreasing *SOX17* expression in the parental cells reduced their sensitivity (Extended Data Fig. 6c,d). These experiments suggest that modulating the activity of a lineage factor can directly and partially affect resistance levels.

Plasticity and EMT are often associated with 'stem-like' states and drug resistance[19], although the association between these two remains poorly understood. One possible model is that the resistance continuum reflects a progressive increase in EMT or stemness, implying a direct explanation for the progressive resistance. Alternatively, EMT or stemness may denote the ability of cells to undergo adaptive phenotypic reprogramming, although not necessarily directly providing high resistance. In this second model, EMT accounts for an essential enabling—although not sufficient—property for subsequent state transitions within the resistance continuum. To distinguish between these two scenarios, we first compared the EMT signature scores across our resistant populations. We observed an early increase in EMT scores within the resistance continuum, yet a progressive increase in resistance did not correspond to higher EMT scores (Extended Data Fig. 6e–i). Next, we experimentally induced EMT in Kuramochi control cells using a lentiviral doxycycline-controlled expression of *Slug* (*SNAI2*), a key regulator of EMT[20]. We selected a Slug overexpression level that conferred enhanced invasion resembling that observed in T320 cells in a Transwell assay (Fig. 2d and Extended Data Fig. 6j–l). We found that the Slug-expressing cells were less sensitive to olaparib compared with their parental controls, highlighting the role of EMT in providing an initial survival advantage. However, these cells were substantially less resistant to olaparib compared to the cells on the resistance continuum. After 3 weeks of exposure to 5 μM olaparib, Slug-overexpressing cells did transition to net-positive growth, although they were notably less resistant than the adapted lines (Fig. 2d). Therefore, these results are consistent with the notion that the EMT programme directly and partially reduces drug sensitivity, while it enables faster adaptation through phenotypic plasticity over longer timescales.

Adapted states are associated with CNAs

We next tested whether the emergence of the adapted cell states is associated with copy number alterations (CNAs). We first inferred CNAs using the scRNA-seq data and found that alterations in the adapted lines largely agreed with our defined state subpopulations (Fig. 3a and Extended Data Fig. 7a), indicating clonal genetic evolution. By contrast, clearly distinguishable CNAs were not present among the persister cells (Extended Data Fig. 7a). We found concordant CNA profiles by performing whole-exome sequencing (WES) (Fig. 3a). We observed significant overlaps between the genes present in the adaptation modules and their gained or deleted status in the CNA regions (Extended Data Fig. 7b). Some CNAs were conserved across the adapted lines (regions of chromosomes 7 and 9), including a deleted region that spans the *CDKN2A/B* locus (validated by WES), thus suggesting a potential contribution to PARPi resistance. We did not detect secondary *BRCA1* or *BRCA2* mutations in the WES data from the resistant lines.

The association between CNAs and cell states may reflect adaptive changes or they may be a consequence of the PARPi treatment, as the ensuing replication stress is expected to induce CNAs[21]. Indeed, we also observed that the dose-escalation experiment across other cell lines (Ovsaho and COV362) led to cell line-specific CNA profiles (Extended Data Fig. 7c), despite exhibiting shared resistance markers (Fig. 2a and Extended Data Fig. 3e,j). Additionally, state S5 cells exhibited distinct profiles (Fig. 3a, see T160 and T320 cells), suggesting that CNAs alone do not completely define an adapted state. This may imply that multiple clones evolved in parallel through the resistance continuum.

Progressive chromatin reprogramming

To gain insight into the epigenetic mechanisms underlying drug adaptation, we measured genome-wide chromatin accessibility by assay for transposase-accessible chromatin with sequencing (ATAC-seq) in the resistant and persister lines. In the adapted lines, we again sorted $CD24^{low}CD44^{hi}$ cells to profile chromatin reprogramming at the subpopulation level (Fig. 3b and Extended Data Fig. 5b). Principal component analysis (PCA) on the resulting chromatin accessibility profiles revealed that PC1 and PC2 together capture the ancestral order from the parental sensitive cells to the T320 adapted cells (Fig. 3c), indicating cumulative changes in chromatin accessibility. Chromatin accessibility from the T5 to the T320 lines revealed a dynamic process of increasing and shared differentially accessible regions (Fig. 3d). By contrast, persister cells exhibited relatively fewer changes (Fig. 3e), mirroring the transcriptome findings (Fig. 2c and Extended Data Fig. 3k,l).

Enrichment analysis of binding sites for transcription factors provided additional support for the widespread epigenetic reprogramming (Extended Data Fig. 8a–d). The adapted lines showed a progressive decrease in accessibility for the lineage drivers SOX17, PAX8 and WT1 (Fig. 3e, left), indicating a coordinated dedifferentiation programme. Acute drug exposure did not induce the same extent of reprogramming in the persister populations. By contrast, the adapted lines exhibited a steady increase in accessibility for the global stress regulators AP1, NRF2 and ATF4 in terms of both the number and proportion of accessible peaks (Fig. 3e, right)—suggesting reinforced activity of stress-response programmes along the olaparib resistance continuum.

We next explored whether the chromatin changes would be stable upon drug removal. After growing the most adapted line for 120 days without drug (T320W), we found that the cells remained resistant to lower doses (10 μM and 40 μM) but exhibited pronounced sensitivity to 320 μM, implying partial reversibility of the resistance (Extended Data Fig. 8e). Notably, we did not observe a tradeoff in proliferation, but rather faster rates in T320 and T320W lines relative to the parental control cells (Extended Data Fig. 8e). Upon drug removal and subsequent short re-treatment, less accessible chromatin regions remained stable, indicating the maintenance of a dedifferentiated state (Fig. 3f, top and Extended Data Fig. 8f). By contrast, the T320W line showed evidence of partial loss of the more accessible chromatin regions that had been gained by their parental T320 line. Upon re-exposure to the drug, however, the T320W line recovered this chromatin accessibility, suggesting an underlying mechanism for cellular memory that reinstates stress-associated regulation (Fig. 3f, bottom and Extended Data Fig. 8f). Collectively, these findings indicate that drug adaptation involves gradual dedifferentiation alongside dynamic recruitment of reinforced stress-responsive regulatory regions.

In vivo adapted and persister states

We next tested whether the adapted and persister states occur in vivo using a *BRCA2*-mutant HGSOC patient-derived xenograft (PDX) model. In a first experiment (hereafter, 'persistence'), we sought to study whether treatment with different doses elicits a persister-like

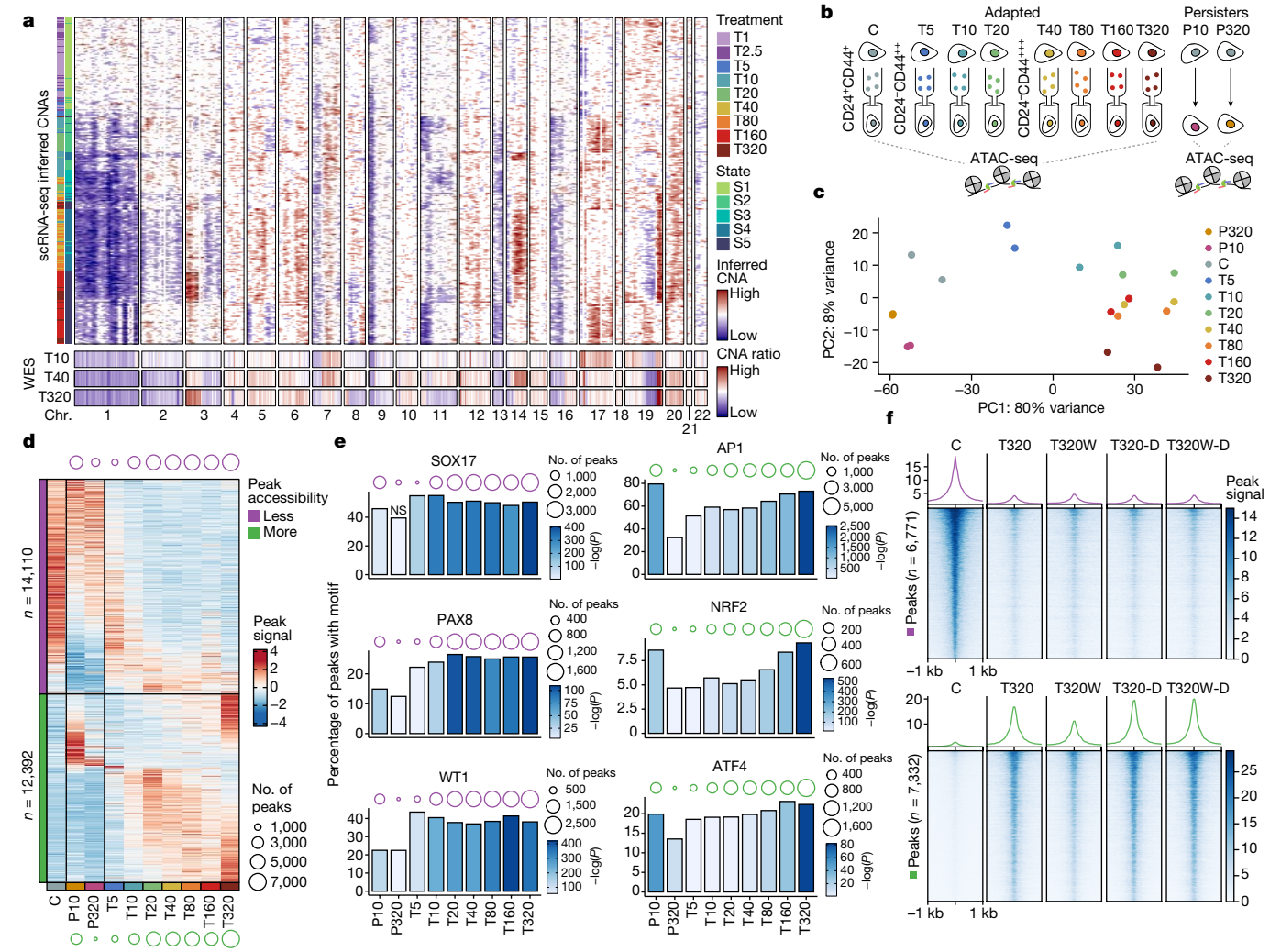


Fig. 3 | CNAs and chromatin reprogramming during drug adaptation. **a**, CNA inference using scRNA-seq data (top) and WES (bottom). Cells (rows) are ordered on the basis of hierarchical clustering. Chr. chromosome. **b**, Schematic of the ATAC-seq data generation for adapted and persister lines. Cell populations from C and T5–T320 lines were sorted on the basis of their CD24 and CD44 profiles (see Extended Data Fig. 5b). Persisters were generated as described in Extended Data Fig. 3k. **c**, PCA on the top 2,000 most variable peaks across conditions. Dots indicate the two replicates for each condition. **d**, Heat map of the ordered differentially accessible peaks relative to untreated cells (C) represented as scaled values of the normalized counts (Methods). The number of less (purple) and more (green) accessible peaks are represented by circle sizes. **e**, Transcription factor motif enrichment analysis. Bar plots represent the percentages of differentially less and more accessible peaks (relative to C) for representative transcription factor motifs. Colour bars represent enrichment *P* values and circle sizes represent the number of peaks (see Extended Data Fig. 8b). **f**, Heat map of the chromatin accessibility signal across the differentially accessible peaks in T320 and T320W without and with drug treatment (D) (relative to C; Fig. 3d). Peak accessibility signal is shown as normalized read coverage at ±1 kb of the peak centre. Histograms summarize the signal across peaks.

state. Cohorts of PDX were treated with the PARPi talazoparib (Fig. 4a) and tumour growth showed the expected dose-dependent responses (Extended Data Fig. 9a). In a second experiment (hereafter, 'resistance'), we tested whether prolonged treatment with a standard dose over three serial passages generates PARPi-resistant tumours with adaptive-like states. Tumours initially showed remission, but relapsed upon treatment interruption. When the treatment was reintroduced, the tumours did not respond, indicating resistance (R1) (Extended Data Fig. 9b). One replicate (R1*) was reimplanted for second (R2) and third (R3) passages of continuous therapy (Fig. 4c), whose tumours failed to respond, indicating stable resistance (Extended Data Fig. 9c,d). scRNA-seq analysis revealed that cells from the persistence experiment were largely intermixed across dose treatments (Fig. 4b). By contrast, cells from the resistance experiment showed greater heterogeneity, with the emergence of resistant-specific clusters primarily consisting of cells from tumours of the second (cluster 7) and third (clusters 8 and 9) passages (Fig. 4d).

Analysis of the differentially upregulated genes revealed that genes from clusters 0 and 6 overlapped highly with genes that defined the untreated Kuramochi cell state, indicating a correspondence between the in vivo and in vitro drug-naive states (Fig. 4e). Genes upregulated in cluster 4 overlap highly with genes that define the persister states (P10 and P320) and module A (Fig. 4e). Genes from cluster 7, mostly composed of cells from R1 tumours, overlap with persister and modules A and E genes, and possibly constitute an intermediate state. Most notably, in the resistance experiment, we found that cells from cluster 9 exhibited gene programmes that overlap significantly with the adaptive states (modules C, D and E; Fig. 4e), providing evidence that the in vitro adapted states also emerge in vivo.

Expression of individual genes further supported the in vitro findings. Specifically, cells from the persistence experiments and certain clusters (5, 6 and 8) from the resistance experiments maintained expression of the lineage markers *SOX17*, *PAX8* and *WT1*. Conversely, their expression was markedly reduced in resistant cells from cluster 9

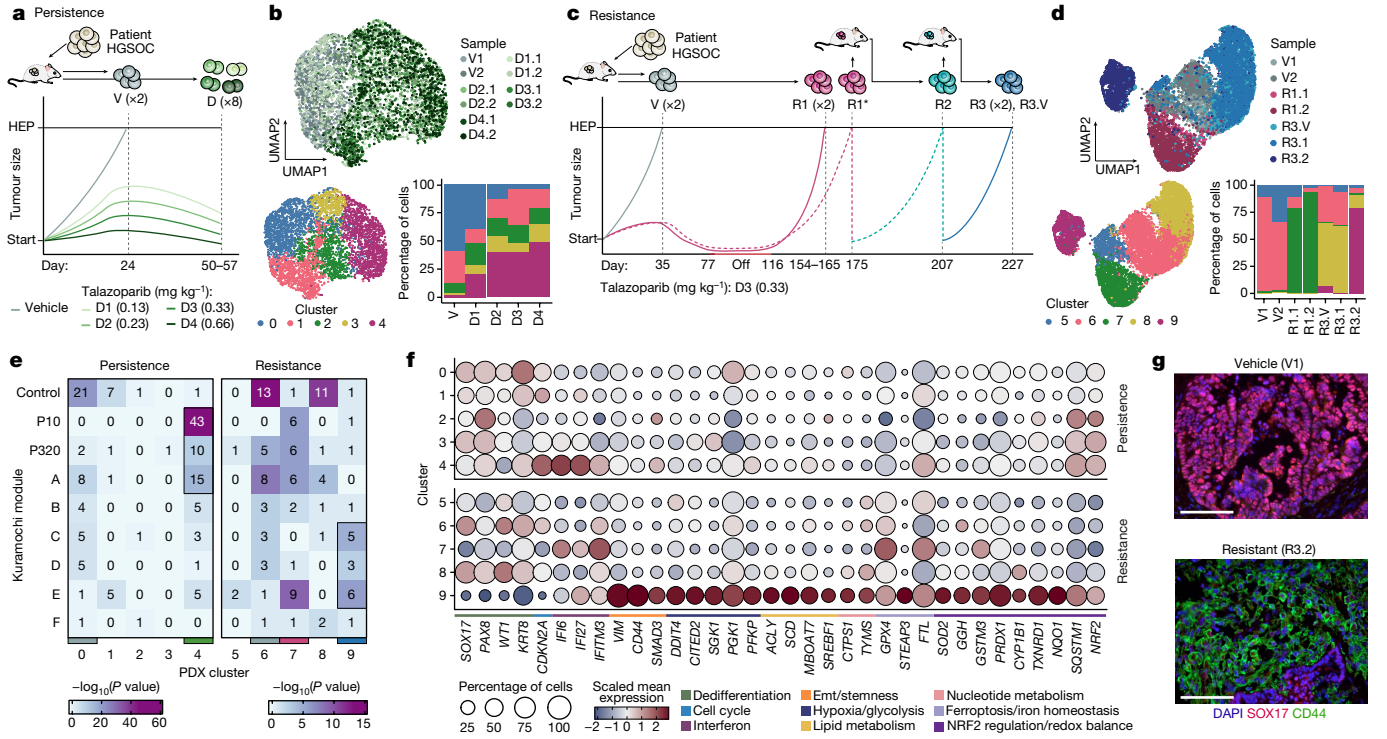


Fig. 4 | In vivo models recapitulate persister and adapted states. a, Schematic of the experimental design for the PARPi (talazoparib) persistence experiment using a patient-derived HGSOC. Two replicates of vehicle (V1 and V2) and dose conditions (D1–D4) were generated. Tumours were collected for scRNA-seq at the humane endpoint (2,000 mm^3) or after 50–57 days of treatment. Tumour growth curves are presented in Extended Data Fig. 9a. **b**, UMAP representation of the clusters for the persistence experiment. Bar plot indicates the percentage of cells from each condition in each cluster. **c**, Schematic of the experimental design for the PARPi resistance experiment. Three passages were performed with talazoparib treatment, with a drug holiday in the first passage (indicated in red). Two replicates of vehicle (V1 and V2) and resistant tumours from passages 1 (R1.1 and R1.2) and 3 (R3.1 and R3.2) (including one treated with vehicle, R3.V) were collected for scRNA-seq at the humane endpoint. A replicate from passage 1 (R1*) was used to seed the second passage (R2). Tumour growth curves are presented in Extended Data Fig. 9b–d. **d**, UMAP and bar plot as described in **b**, but for the resistance experiment. **e**, Gene overlap between the differentially expressed genes across clusters with the in vitro gene modules. Colour scales indicate *P* values (Benjamini–Hochberg corrected) determined from one-sided hypergeometric tests. **f**, Representative differentially expressed genes. Colour bar shows the scaled average expression and circle sizes indicate the fraction of cells expressing the genes. **g**, Immunofluorescence staining of V1 (vehicle) and R3.2 (resistant) tumor sections (scale bars 100 µm, one replicate).

(Fig. 4f and Supplementary Tables 7 and 8), highlighting the role of lineage plasticity in drug adaptation[5,22]. Cells in cluster 9 distinctly express genes associated with the more adapted states, including dedifferentiation/mesenchymal markers (*CD44* and *SMAD3*), metabolic rewiring such as hypoxia/glycolysis (*DDIT4*, *PGK1* and *PFKP*) and fatty acid (*ACLY*, *SCD* and *MBOAT7*) and nucleotide (*CTPS1* and *TYMS*) metabolism (Fig. 4f). Expression of CD44 and SOX17, tested by immunofluorescence, confirmed cell population heterogeneity in the resistant tumour (R3.2) relative to the vehicle and persister tumours (Fig. 4g and Extended Data Fig. 9e).

As observed in the adapted in vitro lines, cells in cluster 9 also exhibited activation of NRF2-dependent processes for antioxidant defence (*NQO1*, *TXNRD1* and *PRDX1*), glutathione metabolism (*GGH* and *GSTM3*) and ferroptosis/iron homeostasis (*GPX4* and *STEAP3*) (Fig. 4f and Supplementary Table 8). Of note, in the persistence experiment, high expression of *NRF2*, its regulator *SQSTM1* (which encodes p62) and oxidative stress genes (*DCXR* and *QSOX1*) showed a dose-dependent trend (Fig. 4f and Extended Data Fig. 9f). Resistant cells in cluster 9 also exhibited loss of *CDKN2A* expression, contrasting with its high expression in non-resistant tumours (cluster 4), indicative of growth arrest[23] (Fig. 4f). Furthermore, cluster 9 cells showed extensive CNAs relative to the persister and other resistant populations (Extended Data Fig. 9g), pointing to a possible role of CNAs in state transitions. These results recapitulate diverse cellular states in a clinically relevant model, with the emergence of adapted states accompanying drug resistance after prolonged treatment.

Systems-level metabolic dependencies

Our findings pointed to metabolic changes related to NRF2-regulated programmes in PARPi-induced oxidative stress. We therefore tested whether the resistance continuum is coupled with a progressive acquisition of systems-level metabolic dependencies. To test this hypothesis, we performed a CRISPR-based negative-selection screen using a metabolic gene-knockout library[24] on Kuramochi drug-naive cells (control cells) and the adapted T10 and T320 lines (including the control cells on 1 µM of olaparib (C_D); and the T320 cells on 40 µM of olaparib ($T320_D$)) (Fig. 5a and Supplementary Tables 9–14). We observed an increasing number of depleted genes between the T10 to T320 when compared to the parental control cells, indicating metabolic reprogramming along the resistance continuum (Extended Data Fig. 10a,b, Supplementary Fig. 9 and Supplementary Tables 15 and 16).

The top depleted genes across the screens performed on the adapted lines T10, T320 and $T320_D$ included NRF2 targets involved in redox balance (*TXNRD1* and *PRDX1*), the NADPH-producing PPP (*PGD*), glycolytic enzymes (*PGK1*, *ALDOA* and *ENO1*), iron homeostasis (*STEAP3*) and nucleotide biosynthesis enzymes (*IMPDH2*, *GUK1* and *DTYMK*) (Fig. 5b and Extended Data Fig. 10a). Specifically to both T320 and $T320_D$, additional nucleotide metabolism enzymes (*TYMS*, *CAD* and *CTPS1*) and genes involved with iron–sulfur cluster biogenesis in mitochondria such as *FDX1* and *FXN* were depleted in these highly adapted cells (Fig. 5b and Supplementary Tables 12 and 13). The adapted $T320_D$ cells were specifically dependent on mitochondrial activity and oxidative

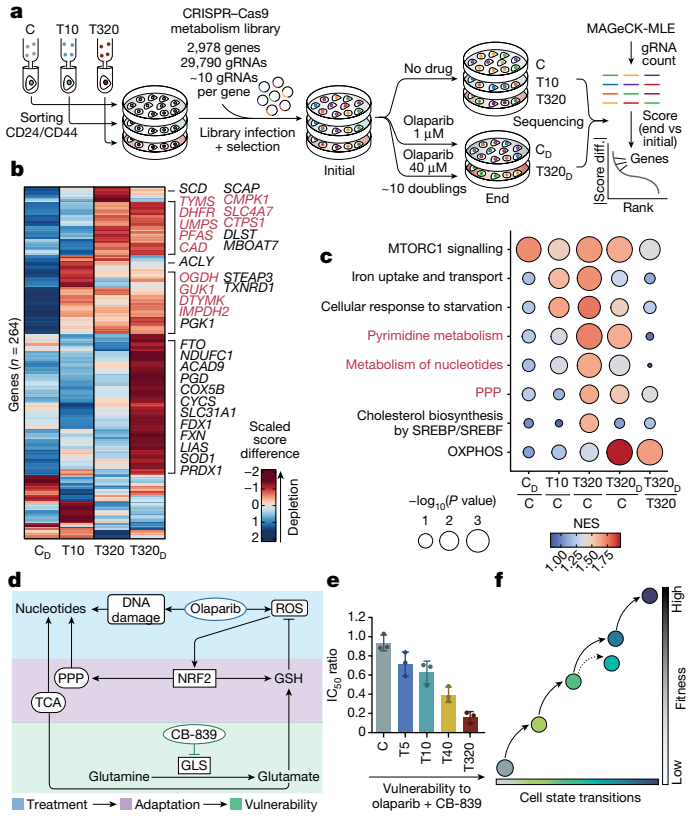


Fig. 5 | CRISPR-based negative-selection screen reveals systems-level metabolic dependencies. a, Schematic of the CRISPR-based screen. Kuramochi cells sorted from C, T10 and T320 lines (see Extended Data Fig. 5b). Score differences (|score diff.|) relative to a reference sample were used to assess gene depletions. gRNA, guide RNA. **b**, Heat map showing depleted genes across conditions relative to C, or C_D and T320 for $T320_D$. Genes were ordered on the basis of hierarchical clustering. Nucleotide metabolism-related genes are shown in red. **c**, GSEA performed on genes ranked by their highest negative score differences relative to C or T320. Selected pathways are shown based on NES and $P < 0.05$ (FDR < 0.25, one-sided Kolmogorov–Smirnov test). Colour scale is the NES as calculated by GSEA and circle sizes indicate *P* values. **d**, Schematic illustrating the rationale for the combined treatment with olaparib and CB-839. GSH, glutathione; TCA, tricarboxylic acid cycle. **e**, Bar plot showing the average of IC_{50} ratios between the combined olaparib and CB-839 treatment relative to olaparib monotherapy (3 independent experiments, 6 technical replaicates). Data are mean ± s.d. **f**, Progressive cell-state transitions along a resistance continuum. Cell states with differential fitness emerge and are selected during adaptation to therapy.

phosphorylation (*COX5B*, *CYCS* and *ACAD9*). Of note, NRF2 targets such as *TKT*, *PRDX1* and *TXNRD1* were even more depleted in T320 cells growing on drug treatment ($T320_D$) when compared with the steady state (Fig. 5b and Supplementary Table 14). Notably, many of these depleted genes are highly expressed in the adaptive gene modules (Fig. 2a and Supplementary Table 1), as well as in the resistant in vivo (Fig. 4f) and other in vitro models (Extended Data Fig. 3e,j), suggesting that they have a functional role in PARPi resistance.

In line with the adaptive modules (Fig. 2a,b), we found, at the pathway level, a dependency on nucleotide metabolism, which increased from T10 to T320 and $T320_D$. Also consistent with the gene expression data, we observed dependencies on genes with a role in the PPP, an antioxidant defence mechanism that produces the reducing co-factor NADPH and generates precursors for nucleotide biosynthesis (Fig. 5c). Targeted metabolomics confirmed an increased abundance of nucleotides as well as their precursors and derivatives in T320 (Extended Data Fig. 10c). Dependencies on cholesterol biosynthesis and oxidative phosphorylation were more specific to T320 and $T320_D$, respectively, whereas T10 and T320 showed dependencies on iron homeostasis and cellular response to starvation pathways (Fig. 5c).

We postulated that cells spanning the resistance continuum would gradually become more susceptible to a drug combination targeting both the emerging dependencies on NRF2 pathways and nucleotide biosynthesis. In lung cancer, hyperactivation of NRF2 pathways leads to an increased dependency on glutaminolysis[25], thus exposing a vulnerability to the glutaminase (GLS) inhibitor CB-839. Ovarian cancer cell lines that highly express GLS have also been shown to be more sensitive to the combination of CB-839 and olaparib. This drug combination decreases the pool of reduced glutathione (a major antioxidant) and suppresses nucleotide biosynthesis, enhancing replication stress[26]. Therefore, chronic DNA damage caused by continuous olaparib treatment may lead to an insufficiency in the nucleotide pool. Upon olaparib-induced reactive oxygen species production, adaptation through activation of NRF2 antioxidant programmes would alleviate reactive oxygen species by regulating glutathione metabolism and PPP, which also provides precursors for nucleotide biosynthesis. The GLS inhibitor CB-839 depletes glutamate, whose availability is important for both glutathione production and nucleotide biosynthesis through tricarboxylic acid cycle intermediates (Fig. 5d).

Testing the combination of olaparib and CB-839, we indeed found that the adapted lines were gradually more vulnerable to the combination; leading to a sixfold decrease in the IC_{50} relative to the olaparib-only treatment in the most adapted line (Fig. 5e and Extended Data Fig. 10d,e). Monotherapy with CB-839 led to only slight sensitivity to CB-839, suggesting a synergistic effect with NRF2 activation (Extended Data Fig. 10f–h).

Discussion

Our study reveals a process that we term the resistance continuum, which involves a gradual adaptation to cancer therapy mediated by a series of cell-state transitions, progressively leading to increasing fitness (Fig. 5f). In our longitudinal in vitro experiments, we leveraged a dose-escalation regime to query the adaptive potential of malignant cells, enabling us to dissect individual stages throughout this continuum.

The series of state transitions characterizing a resistance continuum encompasses axes of dedifferentiation and physiological adaptation. The link between lineage plasticity and partial EMT has been causally associated with resistance[19,27]. Here we propose that whereas EMT induction directly contributes to partial resistance, it more generally facilitates phenotypic plasticity over the long term, enabling the emergence of additional mechanisms that increase drug resistance.

Our work also provides insight into the underpinning of drug adaptation through the assembly of transcriptional programmes and chromatin states, with a steady increase in activity of the stress regulators AP1, NRF2 and ATF4. A possible mechanism is that promiscuous AP1 binding 'pioneers' the enhancer landscape, which is potentiated by recurrent stress stimuli[28,29], thus establishing and maintaining a cellular memory upon continuous stress[30].

The fitness of a cell population may increase by gradual exposure to injury as cells epigenetically 'learn' and fine-tune stress regulatory networks[31], thus priming stable adaptive states. Initially drug-sensitive cells thereby accumulate transcriptional and epigenetic changes upon encountering stress, leading to more robust responses upon subsequent stimulation. We speculate that a similar process may occur during treatment of patients, potentially influenced by drug gradients and tumour spatial heterogeneity[32]. Persister cells, which also rely on stress-related programmes[7,33] and chromatin remodelling[34], may represent an early stage within the resistance continuum. In this scenario, the AP1–NRF2–ATF4 axis may act as a molecular rheostat to support cell survival both in acute treatment and throughout the establishment of

resistance. The precise epigenetic mechanisms that constitute such a cellular memory-based adaptation remain unidentified.

Conceptually, the resistance continuum can incorporate both expression-level and mutational changes. Indeed, a combined effect of both copy-number and expression-level changes was documented in cells that underwent gradual adaptation to different ALK inhibitors[11], or the acquisition of genetic mechanisms evolved from drug-tolerant cells[35]. Mutational processes, including CNAs, might restrict the epigenetic landscapes and accessibility of cell states, thus modifying the trajectories of adaptation.

The spectrum of resistance states may be strongly influenced by pre-existing variability[6,14]. As we demonstrate here, extrinsic forces such as dosing[14] and duration of treatment can shift the adaptive trajectories. It is interesting to ponder whether the mechanisms propelling cells throughout the resistance continuum are primarily shaped by an evolutionary process, stemming from stochastic events, or are deterministically constrained by the underlying developmental programmes. The emergence of states may be seen as variants emerging from historical contingencies[36] and their increased fitness fixed through a selective regime. Conversely, transitions to attractor states[37] induced by therapy may exhibit recurrent properties indicative of a learning process. Both of these processes are likely to be involved in cellular adaptation and it may not be possible to achieve a complete uncoupling.

Finally, we uncovered systems-level dependencies that expose latent vulnerabilities along the resistance continuum. This underscores the importance of monitoring the emergence of adaptive states in clinical practice, as demonstrated in a longitudinal assessment of stress-response states enriched during patient chemotherapy[38]. Given the possibility of pharmacological control over cell-state transitions[39], it might be possible to either impede the adaptive process, or steer the evolving populations toward more treatable fates[40].

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-024-07690-9.

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Methods

Cell culture

Kuramochi cells (*BRCA2* mutant[12]) were cultured in RPMI-1640 (ThermoFisher) with 10% fetal bovine serum (FBS), 1% MEM non-essential amino acids, 1% L-glutamine 200 mM, 1% Antibiotic-Antimycotic solution 100× and 0.25 U ml^{-1} insulin solution (Sigma-Aldrich). Ovsaho (*BRCA2* homozygous deleted[12]) and PC9 (*EGFR* exon 19 deletion) cells were cultured in RPMI-1640 (ThermoFisher) with 10% FBS, 1% L-glutamine 200 mM and 1% penicillin and streptomycin (10,000 U ml^{-1}, ThermoFisher). COV362 (*BRCA1* mutant[12]) and A375 ($BRAF^{V600E}$) cells were cultured in DMEM (ThermoFisher) supplemented with 10% FBS, 1% L-glutamine and 1% penicillin and streptomycin. Cells were maintained at 37 °C with 5% CO_2. Ovarian cancer cell lines were provided by the laboratory of B. Neel. Melanoma A375 cells and lung adenocarcinoma (LUAD) PC9 cells were provided by the laboratories of A. Lund and K.-K. Wong, respectively. The cell line identities were confirmed by genotyping using short tandem repeat fingerprinting and were tested negative for mycoplasma with *Mycoplasma* PCR Detection Kit (ABM).

Dose-escalation experiments

To generate drug-induced resistant populations, 1×10^6 drug-naive cells were initially seeded in 150-mm plates. Twenty four hours after seeding, 1 μM of olaparib (Selleckchem, S1060) was added and cells were maintained under treatment until reaching confluency (>70%), thus characterizing resistance at this dose. Cells were collected (0.25% trypsin/EDTA for 5 min at 37 °C) and a fraction of the surviving population (1×10^6 cells) was seeded again and treated with 2.5 μM olaparib until cells reached confluency. This process was repeated sequentially by doubling the drug concentrations until the cell populations were able to achieve confluency at 320 μM olaparib. The initial dose was determined by cell viability assays (see below), indicating a low starting dose (giving less than 30% of the maximal response (IC_{30})). At each step, aliquots of the adapted populations were frozen (10% DMSO, 50% FBS, 40% medium) for further experiments. Using this approach, we generated nine drug-adapted lines for Kuramochi (T1–T320) and six adapted lines for Ovsaho and COV362 (T1–T40) plus the untreated controls (C). Similar dose-escalation experiments were performed for A375 cells treated with dabrafenib (Selleckchem, S2807), starting at 30 nM until resistant to 4 μM (8 drug-adapted lines plus untreated control), and for PC9 cells treated with osimertinib (Selleckchem, S7297), starting at 8 nM until resistant to 1.2 μM (8 drug-adapted lines plus untreated control).

To compare the resistant phenotypes generated by dose-escalation versus constant dose treatment, we repeated the dose escalation in triplicates for Kuramochi up to 10 μM, while initially drug-naive cells were maintained under treatment with the same doses (1, 2.5, 5 and 10 μM) for the same amount of time in each step of the dose-escalation experiment. Medium with fresh drug was replenished every 3 days. Olaparib was added 24 h after plating, in the final concentrations from a stock solution of 50 mM dissolved in DMSO and maintained at −80 °C. Cell number was determined by trypan blue exclusion (Sigma-Aldrich) and counted on a Countess II cell counter (Life Technologies).

Cell viability assays

Viability experiments to test the resistance phenotypes (here defined as fitness) were performed using the CellTiter-Glo 2.0 Luminescent Cell Viability Assay (Promega) according to the manufacturer's protocol. Before performing the viability assays, frozen aliquots of the resistant cells were thawed and cells were recovered without drug treatment for 7–10 days. Cells were plated in 96-well plates at seeding density of 700 to 800 cells per well. After 24 h, cells were treated with the drug with the same concentrations used in the dose-escalation experiments for 6–9 days for the ovarian cancer cell lines, 4 days for A375 and 5 days for PC9 cells. For the olaparib and CB-839 (MCE, HY-12248) treatment combinations, dose–response curves were generated after 6-day treatments on a range of olaparib doses (1 to 320 μM) with or without 50 nM of CB-839. This was performed on C, T5, T10, T40 and T320 adapted lines after sorting for CD44/CD24 populations as in Extended Data Fig. 5b. The effect of this drug combination was calculated using the Bliss synergy score[41]. DMSO without drug was used as a negative control to normalize the cell viability of drug treated wells. Luminescence was determined using the BioTek Cytation 5 Cell Imaging Multi-Mode Reader (Agilent). Cell viability analysis was performed on GraphPad Prism 9 and IC_{50} values were determined using a nonlinear regression curve fit. Pairwise significant differences between IC_{50} were assessed by Student's two-tailed *t*-tests.

Cell cycle analysis

Cells were treated with 10 μM or 320 μM of olaparib 24 h after plating and cell cycle analysis was performed after 48 h of treatment using propidium iodide staining. Cells were fixed in cold 70% ethanol overnight at 4 °C, washed twice in PBS and resuspended in 1× PBS with 0.2 mg ml^{-1} RNase A (Sigma-Aldrich) and 25 mg ml^{-1} propidium iodide (Biolegend). After 30 min incubation at room temperature in the dark, cells were analysed on a LSR II Flow Cytometer (BD Biosciences). Quantification of cell cycle phases was performed on FlowJo software v.10.8.1 (FLOWJO) using the Watson model.

Drug-tolerant persister generation

Persister cells were generated similarly to those described in refs. 3,42 by seeding 1×10^6 Kuramochi cells in 150 mm plates and treated with either 10 or 320 μM of olaparib for 9 days (greater than 90% inhibition concentration (>IC_{90})). Fresh medium with drug was added every 3 days. For PC9 cells, treatment was 9 days with 1.2 μM of osimertinib. Cells that remained attached to the plates were considered persisters. To confirm that the Kuramochi resistant phenotypes were due to adaptation rather than selection of pre-existing resistant clones, we performed two rounds of treatment-recovery of drug-naive cells. For this, we generated persisters in triplicates with 320 μM (the highest dose used in the escalation experiment) and cells were allowed to recover without drug treatment for 9 days prior to performing viability assays.

Apoptosis assay

To quantify apoptosis and cell viability of drug-tolerant persisters treated with different doses, we generated persister cells with either 10 or 320 μM of olaparib as previously described. The adherent cells were collected (0.25% trypsin/EDTA for 5 min at 37 °C), stained with trypan blue and counted manually using haemocytometer or on a Countess II cell counter. Cells were washed twice in cold PBS and stained with DAPI (ThermoFisher) and Annexin V using the Annexin V Apoptosis Detection Kit (BD Biosciences) according to the manufacturer's instructions. Stained cells were analysed on a LSR II Flow Cytometer (BD Biosciences). Quantification of apoptotic and viable cells was performed on FlowJo software v.10.8.1 (Supplementary Fig. 2).

CD24 and CD44 cell staining and sorting

The untreated and resistant Kuramochi populations were collected and stained with DAPI (D1306, ThermoFisher), anti-CD24-PE (555428) and anti-CD44-APC (559942) (BD Biosciences). Populations were sorted on the basis of their CD24/CD44 profiles using a Sony SY3200 Highly Automated Parallel cell sorter and analysed with BD Facs Diva 8.0.2 and FlowJo v10.8.1. Sorted cells were used either for ATAC-seq or plating for cell viability experiments (Extended Data Fig. 5b).

Invasion assays

Invasion assay was performed in 12 well plates with transparent PET membrane with 8-μm pore inserts (Falcon, 353182). Cells from C, T5, T320, C-$Slug_0$ and C-$Slug_1$ populations were plated in appropriate FBS-free medium containing 0.1% BSA (Sigma) on Transwell insert

membrane covered with 30% Matrigel (BD Biosciences) in PBS. The lower wells contained medium with 10% FBS. Approximately 50,000 cells per well were plated and cultured for 72 h. Matrigel was removed from the Transwell insert and cells that migrated through the membrane were fixed in 100% methanol for 10 min on ice and stained with 0.5% crystal violet solution in 25% methanol. Membranes were cut off and mounted on glass slides with xylene-based Cytoseal mounting medium (ThermoFisher). Slides were scanned at 10× using Jenoptik ProgRes camera.

Live-cell imaging

Cells were plated in a 48 well plate at a low density (6,000 cells per well) and allowed to attach for 24 h. The following day, either DMSO (0.5%) or olaparib (5 μM) was added. Medium and Olaparib or vehicle (DMSO) were subsequently changed every 3 days. Images were taken at least every 24 h. IncuCyte live-cell imaging system (Zoom 2016) was used to collect and analyse images.

PCR

Quantitative PCR reactions were carried out in triplicate on specific genomic regions using SensiFAST SYBR Hi-ROX One-Step Kit (Meridian Biosciences, BIO-73005) on a QuantStudio 7 Pro (ThermoFisher). Primers were: *SNAI2* FW, 5′-TCCTGGTCAAGAAGCATT-3′; *SNAI2* RV, 5′-GAGGAGGTGTCAGATGGA-3′; *RPL19* FW, 5′-GATCCGGAAGC TCATCAAAG-3′ and *RPL19* RV, 5′-GGCTGTACCCTTCCGCTTAC-3′.

Lentiviral overexpression of SOX17 and Slug

Lentiviral vector for constitutive overexpression of SOX17 (TFORF2260) was a gift from F. Zhang (Addgene #141998). Lentiviral vector for inducible expression of *Slug* (Addgene #82162) was Gateway-cloned under a doxycycline induced promoter on a pInducer20 backbone (Addgene #44012). JetPrime reagents and protocol (Polyplus) were followed for transfection into HEK-293T packaging cells. Kuramochi control (C) and adapted lines T5, T40 and T320 were infected with viral lysate before selection in the appropriate antibiotic (puromycin 2 μg ml^{-1} for SOX17; neomycin 2 μg ml^{-1} for Slug). The appropriate selection antibiotic was subsequently added to medium every 3 weeks in culture. To induce Slug, doxycycline was added to the cell medium at the indicated dose for 45 days prior to experiment initiation.

shRNA knockdown of SOX17

SOX17 inducible lentiviral small hairpin RNA (shRNA) gene set was purchased from TransOmic Technologies (TLHSU2330). JetPrime reagents and protocol (Polyplus) were followed for transfection into HEK-293T packaging cells. Kuramochi Control cells were infected with one of three shRNA constructs or a non-targeting scrambled shRNA negative control construct by viral lysate before selection in the appropriate antibiotic (puromycin 1.5 μg ml^{-1}). To induce *SOX17* knockdown, doxycycline (500 ng ml^{-1}) was added to the cell medium for 3 days prior to drug exposure.

Generation of patient derived xenografts and drug treatments

All animal experiments were carried out under an approved animal protocol by the Institutional Animal Care and Use Committee (IACUC) of NYU Langone Grossmann School of Medicine. Immunodeficient six-week-old female NOD SCID gamma (NSG) (NOD.Cg-*Prkdc*^{scid} *Il2r*^{gtm1Wjl}/*SzJ*) mice were purchased from Jackson Laboratory and allowed to acclimatize to housing conditions. Animals were housed in an specific-pathogen-free-grade animal facility at 25 ± 2 °C with a humidity of 40–70%. Besides, 12-h:12-h light periods were alternated between day and night, and these mice were given an unrestricted diet. Before the experiment, these mice were adaptively reared for more than three days. To generate high-grade serous ovarian cancer (HGSOC) PDX models of PARPi treatments, we used a previously characterized *BRCA2*-mutated tumour derived from PDX models[43] (ID: 66799). Approximately 1×10^6 cells in a 1:1 ratio with Matrigel Matrix (Corning) were subcutaneously injected with a 27-gauge needle (BD Micro-Fine) into the right lower mammary fat pad on the abdominal dorsal surface. Observation of animals including caliper measurements of tumour volume and body weight continued twice a week. Mice were euthanized with CO_2 (confirmed by cervical dislocation) and tumours were excised for further procedures at sizes not exceeding the IACUC permitted measurements or volumes (up to 2,000 mm^3 or up to 20 mm in one dimension). When tissue was available, a portion of the tumours was embedded in optimal cutting temperature and snap-frozen in a bath of isopentane chilled over liquid nitrogen. The tumour blocks were stored at −80 °C until use for staining. Tissue sections were mounted onto a slide at a thickness of 5–7 μm. Haematoxylin and eosin staining was performed and annotated by an expert gynaecologic pathologist.

PARPi dose–response. To study the transcriptional changes that occur upon different dose regimen, the *BRCA2*-mutant HGSOC cells were injected as previously described and mice were randomized into five groups after reaching 100–200 mm^3 tumour volume. Treatments were performed as follows: (1) Vehicle controls (10% dimethylacetamide, 5% Solutol HS15, 85% PBS) ($n = 2$); (2) dose 1 (D1) was 0.13 mg kg^{-1} of talazoparib (Selleckchem, S7048) ($n = 2$); (3) dose 2 (D2) was 0.23 mg kg^{-1} ($n = 2$); (4) dose 3 (D3) was 0.33 mg kg^{-1} ($n = 2$); and (5) dose 4 (D4) was 0.66 mg kg^{-1} (5 consecutive days a week, oral gavage). Mice were monitored by routine palpation and tumour sizes measured by caliper. Vehicle tumours were collected at sizes not exceeding the humane endpoint and talazoparib treated tumours were collected after 50–57 days of treatment.

PARPi resistance model. To establish a drug-resistance model, tumours were treated along three passages. In a first passage, when the PDX tumours reached 100–200 mm^3 tumour volume, vehicle (10% dimethylacetamide, 5% Solutol HS15, 85% PBS) or talazoparib (0.0066 mg per mouse = 0.33 mg kg^{-1}, oral gavage[44]) was administered for 5 consecutive days a week. Talazoparib treatment caused tumour remission and treatment was interrupted. Tumour sizes were followed until reappearance and treatment was reintroduced. Vehicle and drug treated tumours that did not show further regression upon re-treatment—thus indicating resistance—were collected for scRNA-seq at sizes not exceeding the humane endpoint. A replicate tumour was implanted in a second passage and maintained under treatment upon tumour detection. The resulting tumour was implanted in a third passage, generating three replicates (two talazoparib treated and one vehicle).

Tumour dissociation for scRNA-seq

Upon arrival to the laboratory, tumours were minced with a razor blade and enzymatically dissociated in two steps. First, minced samples were incubated with 5 ml of digestion mix 1 (9 ml Advanced DMEM/F12, 1 ml Gentle Collagenase and 100 μl DNase I) at 37 °C for 30 min, centrifuged at 1,000 rpm for 4 min. After removing the supernatant, samples were incubated with 5 ml of digestion mix 2 (3 ml Advanced DMEM/F12, 2 ml Dispase, 5 ml TrypLE, 100 μl DNase I) at 37 °C for 15 min, placed on a 70-μm cell strainer and washed with Advanced DMEM/F12 + 10% FBS. Samples were centrifuged at 1,000 rpm for 4 min, incubated with 1 ml of ACK Lysing Buffer (ThermoFisher) for 4 min, washed with PBS + 10% FBS, resuspended in PBS, placed on a 35-μm cell strainer and viability was determined by trypan blue staining, which ranged from 65% to 90%. Dissociated cells were collected for scRNA-seq immediately (resistance experiment) or cryopreserved (dose–response experiment) by resuspending in freezing medium (50% FBS, 40% Advanced DMEM/F12 and 10% DMSO) and stored at −80 °C. Batches of replicates were processed on the same day. Frozen samples were thawed, washed in medium, resuspended in PBS and the cell concentration adjusted for the 10× Genomics single-cell encapsulation protocol following library preparation and sequencing.

Tissue staining

Serial sections following haematoxylin and eosin staining were used for immunofluorescence experiments. Following tissue retrieval with acetone, tissues were rehydrated in PBS, and fixed in a solution of 1.6% paraformaldehyde. Sections were blocked and permeabilized for 1 h in a solution of 3% BSA in PBS with 0.1% Triton X-100. Sections were stained with primary antibodies for SOX17 (R&D Systems AF1924-SP, 10 μg ml^{-1}) and CD44 (R&D Systems MAB7045-SP, 10 μg ml^{-1}) for 3 h at room temperature in 3% BSA. Sections were subsequently stained with secondary antibodies (ThermoFisher Scientific, rabbit anti-mouse, A-11059, 1:200) and (ThermoFisher Scientific, donkey anti-goat, A-21432, 1:200), followed by DAPI (1 μg ml^{-1}). Sections were mounted with Gold Antifade Mountant solution (Invitrogen, P36934) and imaged using a Keyence BZ-X Series microscope. All immunofluorescence staining was performed in triplicate for each tumor sample, and representative images of one replicate per tumor were selected.

inDrop library preparation and sequencing

Cells from the Kuramochi dose-escalation experiment (C and T1–T320) were collected when they reached confluency at their respective dose treatments and collected for single-cell encapsulation and library preparation using the inDrop platform[45] as previously described[46]. Libraries were sequenced on an Illumina NextSeq 500 and reads were processed using a custom inDrop pipeline as described[46] using the hg38/GRCh38 human genome assembly and the Ensembl 93 transcriptome annotation.

10X Genomics library preparation and sequencing

We used the 10X Genomics workflow for scRNA-seq on cells collected for the comparison between Kuramochi persisters (P10 and P320) and adapted (T10 and T320) populations, as well as for Ovsaho, COV362, A375, PC9 dose-escalation experiments and for PDX data. Cell loading for droplet-based scRNA-seq and library preparation was performed using the 10X Genomics Chromium platform with the Chromium Single Cell 30 Library & Gel Bead Kit v3.1 following the manufacturer's instructions. Libraries were diluted to 2 nM and 75 bp paired-end sequencing was performed using the Illumina NextSeq 500 and 150-200 million paired reads were generated for each library. Raw reads were processed with the CellRanger v3.1 or v.7.0.0 (10X genomics) pipeline (default parameters) with the hg38/GRCh38 genome and Ensembl 93 transcriptome references. For PDX samples, reads were processed with the CellRanger multi-species detection and only cells assigned to 'human' were considered for further analysis.

Cell multiplexing for pooled scRNA-seq. To compare the expression among untreated (C), persister (P10 and P320; two replicates) and the adapted cells (T10 and T320), we used the cell hashing method[47] to tag cells from different conditions based on the expression of surface markers (TotalSeqA, Hashtag oligonucleotides (HTOs) 1–5, Biolegend). Five samples from each condition were pooled into a single-cell suspension in equal amounts and loaded in the 10X Genomics Chromium Controller with Single Cell 3′ v3.1 system and libraries prepared according to the manufacturer's protocol. For the hashtag oligos, 1 μl of HTO PCR additive primer was added to the 10X cDNA amplification step, and the supernatant from the 0.6× cDNA cleanup was kept and processed according to the Cell Hashing protocol, with 14 PCR cycles and a 1.2× cleanup after the PCR. The same procedure was applied to pool cells from Ovsaho (C and T5–T40; HTOs 1–5) and COV362 (C and T5–T40; HTOs 6–10) experiments. Sample demultiplexing was performed using the CITE-seq-Count pipeline (v1.4.3; https://github.com/Hoohm/CITE-seq-Count) and Seurat[48]. For the A375 and PC9 scRNA-seq, cell suspensions from each corresponding sample were labelled using the 10X Genomics 3′ CellPlex kit Set A according to the manufacturer's instructions. Cells were pooled in equal amounts and loaded in the 10X Genomics Chromium Controller. Demultiplexing of the corresponding CMOs was performed using CellRanger v.7.0.0 with 0.8 as assignment confidence.

ATAC-seq sample preparation and sequencing

ATAC-seq samples (C, P10, P320, T5–T320 and T320W) were prepared in duplicates using a modified version of the OMNI-ATAC protocol[49]. Briefly, cells grown in tissue culture under drug treatment with their respective adapted doses were trypsinized, stained for expression of CD44 and CD24 and sorted as previously described (C and T5–T320). To assess the stability of the chromatin states, T320 cells were grown for 120 days without drug treatment, generating T320W. Together with its parental T320 (grown without drug for 10 days), cells were re-exposed to 320 μM. Cells from these four conditions (T320 and T320W on and off drug treatment) were collected for ATAC-seq in duplicates. Briefly, approximately 50,000 cells per condition were used for nuclei extraction using the ATAC-seq resuspension buffer as previously described[49]. Nuclei were resuspended in 50 μl of transposition mix (25 μl 2× TD buffer, 2.5 μl transposase (100 nM final), 16.5 μl PBS, 0.5 μl 1% digitonin, 0.5 μl 10% Tween-20, and 5 μl water). Transposition reactions were incubated at 37 °C for 40 min in a thermomixer with shaking at 1,000 rpm. Reactions were cleaned up with Zymo DNA Clean and Concentrator 5 columns. ATAC-seq libraries were amplified using Ad2_noMx (universal forward) and Ad2.x (indexed reverse). Libraries were diluted to 2 nM and 75 bp paired-end sequencing was performed using the Illumina NextSeq 500/550. Between 50 and 200 million paired reads were generated for each library.

Single-cell data analysis

Filtering. Raw gene expression matrices were TPM (transcripts per million)-normalized (scale factor of 10^4) and $\log_2$-transformed. For each cell, the number of transcripts, genes expressed, and the proportion of transcripts derived from mitochondrial and ribosomal genes were determined. For each dataset, cells with transcript number lower than 500 or higher than 2× s.d., with less than 500 genes detected and with higher than 20% of transcripts corresponding to mitochondrial genes were excluded from further analysis. Genes expressed in less than 1% of the cells in each dataset were removed.

Cell clustering and differential gene expression. For all samples we observed that differences due to cell cycle gene expression were a major source of variation. To mitigate this effect for the interpretation of cell states, we used the Seurat function CellCycleScoring to assign cells to G1, S and G2/M phases. We then considered the G1 cells for further analyses. Dimensionality reduction, clustering and differential expression were performed using the Seurat R package v4.0.5 and v2.3.2. To control for unwanted sources of variation, the number of unique molecular identifiers (UMIs) and the fraction of mitochondrial and ribosomal UMIs were regressed out during data scaling. For the individual analysis of the Kuramochi datasets (C-T320) we used the DrImpute method (k parameter = 5:10)[50] before clustering. Highly variable genes were identified using the Seurat FindVariableGenes function (default parameters) and these were used to compute the PCA following UMAP using the top five to ten principal components (cutoffs determined by the elbow method). Louvain clustering was performed using the FindClusters function using the selected pricipal components with resolution set to 0.3 to 0.6. For the analysis of PDX dose–response samples, we observed a batch effect driven mostly by one of the vehicle samples and we used the fastMNN method[51] for correction of the aggregated datasets before clustering. Differential gene expression across clusters was performed using the FindMarkers function using the MAST test (logfc.threshold=0.25, min.pct=0.2, min.cells.genes=10).

Defining cell states across resistant populations. To study cellular states across the Kuramochi resistant populations, we first assessed the

differentially expressed genes among the cell clusters within each sample separately. The top 100 most differentially expressed genes (ranked by their adjusted *P* values) in each cluster were selected to generate a non-redundant list of genes (n = 1,511). These genes were used to build an average expression matrix of each cell cluster across all samples. A correlation matrix (Spearman's correlation coefficients) across these average profiles was used for average linkage hierarchical clustering and the cell states were defined based on the cluster structure (k = 5).

Gene module identification. To identify gene expression programmes across the Kuramochi lines from C to T320, we implemented the approach of Neftel et al.[52], with some modifications. First, we aggregated all G1 cells (non-imputed dataset) from all conditions and performed an average linkage hierarchical clustering using the 1,000 most highly variable genes (defined by the FindVariableGenes function) on the centred dataset using one minus the Pearson correlation as the distance metric, as proposed previously[52]. To evaluate all possible clusters of cells that have a potential gene expression signature, we recovered all clusters excluding by size (less than 30 cells and no more than 80% of all cells) and by signal of differential expression. Each cluster of cells was evaluated for a signature of differential expression relative to the remaining cells using the FindMarkers function as described above. Clusters with a signature of differentially expressed genes were retained according to the following criteria: genes expressed >1.8 average fold change, adjusted *P* value < 0.001 and with more than 50 genes matching these criteria. Gene signatures were then hierarchically clustered based on their Jaccard index and, for pairs with similarity above 0.75, the signature with the largest number of genes was considered. To integrate the gene signatures into gene modules, clusters with less than three signatures were excluded. The genes from all signatures that constitute a particular cluster were merged and ranked based on the ratio of their average expression in cells from a particular cluster relative to their average expression in cells from the other clusters. The gene module definition did not enforce mutual exclusivity, allowing genes to be associated with multiple modules. To assess the significance of gene overlap across modules and differentially expressed gene lists from other analyses, *P* values were computed with Fisher's Exact tests implemented in the GeneOverlap v.1.26 R package. To assign EMT scores, cells were scored for the Hallmark EMT gene set (MSigDB) using the UCell 2.0.0 package in R. To assess the significance of overlap among genes in modules and genes in CNA regions, permutation tests (10,000) were performed using the regioneR 1.10.0 package[53].

Identification of genes with dose–response patterns. Using the PDX datasets from the dose–response experiments, we identified genes with dose–response patterns by first performing pairwise differential expression across all treated conditions versus vehicles for each batch of replicates independently. Genes differentially expressed in both of the replicates were considered. A non-redundant list of these differentially expressed genes was used to generate an average expression matrix across conditions. The samples were ordered from low to high doses (V, D1, D2, D3, D4) and the genes were clustered using the Mfuzz 2.5.0 package[54] setting the number of clusters to 7 and m parameter = 1.8. Genes with membership score higher than 0.3 were included and those with similar profiles of decreasing or increasing expression relative to high (or low) doses in both replicates were aggregated.

Single-cell inferred CNAs

Large-scale chromosomal alterations from scRNA-seq datasets were inferred by the inferCNV v1.4.0 package. Untreated samples were used as a reference for CNA inference in the treated samples. The default options were used except for a cut-off of 0.1 for the minimum average read counts per gene and clustering by groups was set based on the cluster annotations determined from the transcriptome analyses. The output consisting of a denoised modified expression matrix as genes ordered by chromosomal location was used for heat map visualization considering 30% of the cells from each condition with ComplexHeatmap 2.11.1 in R.

Gene set enrichment analysis

Gene set enrichment using the hallmark annotations (MSigDB), KEGG and Reactome pathways was performed using the R package fgsea v1.16. Differential expression between groups was determined using the MAST algorithm implemented in Seurat and the genes were ranked using the sign of the average log fold change multiplied by the $-\log_{10}$ of *P* value. Ranked lists were used as inputs for the fgsea package with a minimum gene set size of 10 and a maximum of 500, and significance assessed by 10,000 permutations. Terms were considered significantly enriched at *P* values < 0.05 and FDR < 0.25 as suggested in the GSEA manual. For the analysis shown in Fig. 2b, the comparison was performed across the adapted cells (that is, excluding untreated cells that belong to the module A annotation).

ATAC-seq data processing and analysis

Peak detection. Adapter sequences were removed using TrimGalore 0.5.0 (https://github.com/FelixKrueger/TrimGalore) (-q 20, - --length 30). Filtered reads were aligned to the human genome (hg38/GRCh38) with Bowtie2[55] v2.3.4.1 (--very-sensitive, -k 10, -X 2000). After removing reads mapping to mitochondrial DNA and duplicate reads removed with PicardTools 2.18.2 MarkDuplicates (http://broadinstitute.github.io/picard/), only properly mapped pairs assigned by SAMTools 1.9[56] were included for further analysis. To avoid peak detection bias due to sequencing depth differences, read depth across samples was downsampled with SAMTools to match the lowest depth. Peak calling was performed with MACS2 v2.2.4[57] and blacklisted regions were excluded[58]. Reproducible peaks between replicates for each sample were determined using the IDR 2.0.3 framework (https://github.com/nboley/idr) and peaks with high reproducibility (IDR score <0.05) were considered. A consensus peak set across all samples was constructed by merging the reproducible peaks from each sample using HOMER v4.1[59]. Finally, a read count table for the consensus peaks was generated using deepTools 3.1[60].

Differential peak accessibility and motif analysis. The count table of consensus peaks was normalized and differentially accessible peaks were calculated using DESeq2 1.18.1[61]. Differentially accessible peaks relative to the untreated sample (C) were considered as adjusted *P* < 0.001 and fold change ≥ 3. The top 2,000 most variable peaks were used for PCA. To visualize the peak signal (read counts) across samples, Zavit[62] was used to sort the mean peak signal (between replicates) across samples. Motif enrichment analysis on the differentially accessible peaks was performed using HOMER's findMotifsGenome for known transcription factor motifs. To assess the significance of the overlap between the genomic regions of the differentially accessible peaks and the regions that encompass CNAs (determined by WES), permutation tests (1,000 permutations) were performed using the regioneR package. To verify the influence of CNAs on the motif enrichment patterns, the motif analysis was repeated by removing the more or less differentially accessible peaks that overlap with gained or deleted copy-number regions, respectively. To assess the stability of the chromatin states in T320 after drug removal (and re-exposure, Fig. 3f), the normalized peak signal (±1 kb of the peak centre) of the differentially accessible peaks for T320 cells (versus C) was determined using deepTools (bamCoverage --binSize 1, --normalizeUsing RPGC; computeMatrix --binSize 1, --referencePoint center) and visualized with the plotHeatmap command. Differences in peak signal were assessed with Wilcoxon tests.

Whole-exome sequencing and analysis

Genomic DNA extraction for WES was performed using Qiagen Blood and Cell DNA kit (13323). The extracted DNA from representative

samples (C, T10, T40 and T320) was sent to the Broad Institute facility and processed according to the in-house pipeline. Briefly, 125 ng in 50 µl was prepared for sequencing using the KAPA Hyper Prep Kit. Hybridization and capture were performed using the relevant components of IDT's XGen hybridization and wash kit and following the manufacturer's suggested protocol. Sequencing was performed using Illumina Novaseq.

Sequencing results were demultiplexed and converted to FASTQ format using Illumina bcl2fastq 2.2.0 software. The FASTQ files were processed using the Seq-N-Slide v21.1 pipeline (https://doi.org/10.5281/zenodo.5550459). Adapter sequences were trimmed with Trimmomatic 0.36[63] and then aligned to the human reference genome (build hg38/GRCh38) using the Burrows–Wheeler Aligner with the BWA-MEM algorithm[64]. Low confidence mappings (mapping quality <10) and duplicate reads were removed using Sambamba 0.6.8[65]. Further local indel realignment and base-quality score recalibration were performed using the Genome Analysis Toolkit (GATK 3.8.1)[66]. Copy-number profiles were calculated using Control-FREEC[67] with untreated samples as the matched controls. ANNOVAR (170716)[68] was used to annotate variants with genomic context such as functional consequence on genes and identify presence in public variant databases gnomAD 3.0 and COSMIC 90.

Metabolic CRISPR–Cas9 genetic screen

The human metabolic CRISPR–Cas9 knockout library was a gift from K. Birsoy[24]. Virus was produced by cotransfection of HEK293 cells with lentiviral library pool with packaging vectors (psPAX2 and VSV-G) using PEI Pro (PolyPlus). The titre of lentiviral supernatants was determined by infecting target cells with several amounts of virus in the presence of polybrene (8 µg ml^{-1}; Millipore), counting the number of puromycin-resistant infected cells after 3 days of selection. Kuramochi C, T10 and T320 (CD44^{hi}CD24^{low}, sorted as in Extended Data Fig. 5b) cells were infected at a multiplicity of infection of ~0.3 and selected with puromycin (4 µg ml^{-1}) 48 h after infection. An initial pool of cells was collected for genomic DNA extraction (initial time point). Remaining cells were cultured for 10 population doublings either off (C, T10, T320) and on olaparib treatment (C in 1 µl and T320 in 40 µl), after which cells were collected for genomic DNA extraction (end time point). sgRNA inserts were PCR-amplified, gel purified, quantified with NEBNext Library Quantification Kit (E7630S) and sequenced on a Illumina NextSeq 500/550, generating between 20–40 million paired reads for each library.

Data analysis of the metabolic CRISPR–Cas9 genetic screen. Sequencing reads were mapped to the sgRNA library sequences with MAGeCK-count[69] algorithm implemented in MAGeCK 0.5.8, resulting in 75–80% of mappability. The sgRNAs counts were used as inputs for the MAGeCK-MLE algorithm to calculate gene scores, which takes into account the differences in sgRNA abundance from the initial and the end points of the screen. Genes were considered depleted in a sample when having negative β scores, $P < 0.05$ and FDR < 0.25, as suggested in the MAGeCK manual. To analyse gene depletions across samples (Fig. 5d), only genes with negative β scores were considered. Gene score differences between a sample and a reference were determined considering depleted genes (negative β scores in the sample) with FDR < 0.25 and negative score differences higher than one standard deviation between the sample and the reference. The samples C_D, T10 and T320 were compared relative to C, and the $T320_D$ included comparisons with T320 and C_D. The overlap among depleted genes in a sample relative to C were computed with Fisher's Exact tests implemented in the GeneOverlap R package (Fig. 5c). For GSEA analysis, genes were ranked based on the negative score differences between a sample and a reference (for example, T320 versus C). Only genes with negative score differences were considered. GSEA was performed using the fgsea R package and significant terms were selected with $P < 0.05$ and FDR < 0.25.

Immunoblotting

Samples were collected and lysed in the Laemmli buffer. Protein concentrations were determined using the Bio-Rad DC Protein Assay. Twenty milligrams of total protein was separated on a 4 to 12% bis-Tris gradient gel (Invitrogen) and then transferred onto PVDF membrane (Bio-Rad). Proteins were immunoblotted with the indicated antibodies. Goat anti-Rabbit immunoglobulin G (IgG) (H + L) horseradish peroxidase (HRP) (31460) and Goat anti-Mouse IgG (H + L) HRP (31430) were used as secondary antibodies (Invitrogen, ThermoFisher Scientific). Immunoblots were visualized (Amersham Imager 600 or 680) using the Pierce ECL Western Blotting Substrate (Thermo Scientific, 32106), and quantification was performed using ImageJ software. Anti-beta actin antibody (Santa Cruz, sc-47778 or mAbcam 8226) was used as a loading control for all of the following staining, the NRF2 (Cell Signaling Technology (CST), 12721), ATF4 (CST, 11815), HO-1 (HMOX1, MA1-112), Slug (CST, 95855) and SOX17 (CST, 8177). Full scans are provided in Supplementary Figs. 6–8 and 10.

Targeted metabolomics profiling

Sample processing. Kuramochi cells (C and T320) were cultured until ~80% confluence either on 320 µM of olaparib for 24 h or off treatment in triplicates per condition. Cell pellets were collected and frozen with liquid nitrogen for submission to the NYU Metabolomics Core Resource Laboratory. Samples were analysed by liquid chromatography–mass spectrometry to detect and quantify known peaks. A metabolite extraction was carried out on each sample based on a previously described method[70]. The LC column was a MilliporeTM ZIC-pHILIC (2.1 × 150 mm, 5 µm) coupled to a Dionex Ultimate 3000TM system and the column oven temperature was set to 25 °C for the gradient elution. A flow rate of 100 µl min^{-1} was used with the following buffers: 10 mM ammonium carbonate in water, pH 9.0 (buffer A) and neat acetonitrile (buffer B). The gradient profile was as follows: 80–20% B (0–30 min), 20–80% B (30–31 min), 80–80% B (31–42 min). Injection volume was set to 2 µl for all analyses (42 min total run time per injection). Mass spectrometry analyses were carried out by coupling the liquid chromatography system to a Thermo Q Exactive HFTM mass spectrometer operating in heated electrospray ionization mode. Method duration was 30 min with a polarity switching data-dependent top 5 method for both positive and negative modes. Spray voltage for both positive and negative modes was 3.5 kV and capillary temperature was set to 320oC with a sheath gas rate of 35, aux gas of 10, and max spray current of 100 µA. The full mass spectrometry scan for both polarities utilized 120,000 resolution with an AGC target of 3×10^6 and a maximum IT of 100 ms, and the scan range was from 67–1,000 m/z. Tandem mass spectra for both positive and negative mode used a resolution of 15,000, AGC target of 10^5, maximum IT of 50 ms, isolation window of 0.4 m/z, isolation offset of 0.1 m/z, fixed first mass of 50 m/z, and 3-way multiplexed normalized collision energies (nCE) of 10, 35, 80. The minimum AGC target was 10^4 with an intensity threshold of 2×10^5. All data were acquired in profile mode.

Data analysis. The resulting ThermoTM RAW files were converted to mzXML format using ReAdW.exe version 4.3.1 to enable peak detection and quantification. The centroided data were searched using an in-house Python script, Mighty_skeleton version 0.0.2 and peak heights were extracted from the mzXML files based on a previously established library of metabolite retention times and accurate masses adapted from the Whitehead Institute[71], and verified with authentic standards and/or high resolution MS/MS spectral manually curated against the NIST14MS/MS[72] and METLIN (2017)[73] tandem mass spectral libraries. Metabolite peaks were extracted based on the theoretical m/z of the expected ion type—for example, $[M + H]^+$, with a ±5 ppm tolerance, and a ±7.5 s peak apex retention time tolerance within an initial retention time search window of ±0.5 min across the study samples. The resulting data matrix of metabolite intensities for all samples and blank controls

was processed with an in-house statistical pipeline, Metabolyze version 1.0, and final peak detection was calculated based on a signal to noise ratio (S/N) of 3× compared to blank controls, with a floor of 10,000 (arbitrary units). For samples where the peak intensity was lower than the blank threshold, metabolites were annotated as not detected, and the threshold value was imputed for any statistical comparisons to enable an estimate of the fold change as applicable. The resulting blank corrected data matrix was then used for all group-wise comparisons, and *t*-tests were performed with the Python SciPy (1.1.0) library to test for differences and generate statistics for downstream analyses. Any metabolite with $P < 0.05$ was considered significantly regulated (up or down). Heat maps were generated with hierarchical clustering performed on the matrix values utilizing the R library ComplexHeatmap.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Raw and processed data from this manuscript have been deposited in the NCBI Genome Expression Omnibus (GEO) under the accession number GSE247691. Metabolomics data have been deposited in the Metabolomics Workbench (study ID: ST003102; https://doi.org/10.21228/M8WM7T).

Code availability

Relevant code used in this study is available at https://github.com/yanailab/drug-resistance.

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Author contributions G.S.F. conceived the study with I.Y., and performed the experiments and all data analyses. M.B. assisted with scRNA-seq experiments. B.R.K., S.M., D.A. Levine and T.L. designed the in vivo resistance experiments and B.R.K. performed scRNA-seq for these experiments. B.R.K., S.M., G.S.F., T.L. and I.Y. designed the persistence in vivo experiments. B.R.K., G.S.F. and M.P. performed scRNA-seq for these experiments with assistance from D.B. S.M. performed all of the mouse work. M.P. performed and assisted with in vitro experiments. G.S.F., J.P.B., T.P. and I.Y. designed the CRISPR screen experiments. G.S.F. and J.P.B. performed the CRISPR screen experiments under the supervision of T.P. and I.Y. G.S.F., A.B., A.M. and I.Y. designed the SOX17 and Slug functional experiments. A.B. performed these experiments under the supervision of A.M. G.S.F. and A.R. designed and performed the ATAC-seq, scRNA-seq and WES experiments. I.D. contributed with WES data processing and analysis. G.S.F. and A.S.P. performed additional scRNA-seq and ATAC-seq experiments. K.H.T., M.C. and A.S.P. assisted with in vitro experiments. D.A. Liberman assisted with data analysis and staining. G.A. and F.K. assisted with scRNA-seq. G.S.F and I.Y. wrote the manuscript with inputs from A.M. I.Y. oversaw the project. All authors commented on the manuscript.

Competing interests D.A. Levine is a founder of Resident Diagnostics and is a full-time employee of Merck & Co. T.P. received funding from Dracen Pharmaceuticals, Kymera Therapeutics, Bristol Myers Squibb and Agios Pharmaceuticals not related to the submitted work. T.P. consulted for Vividion Therapeutics, Tohoku University and Faeth Therapeutics.

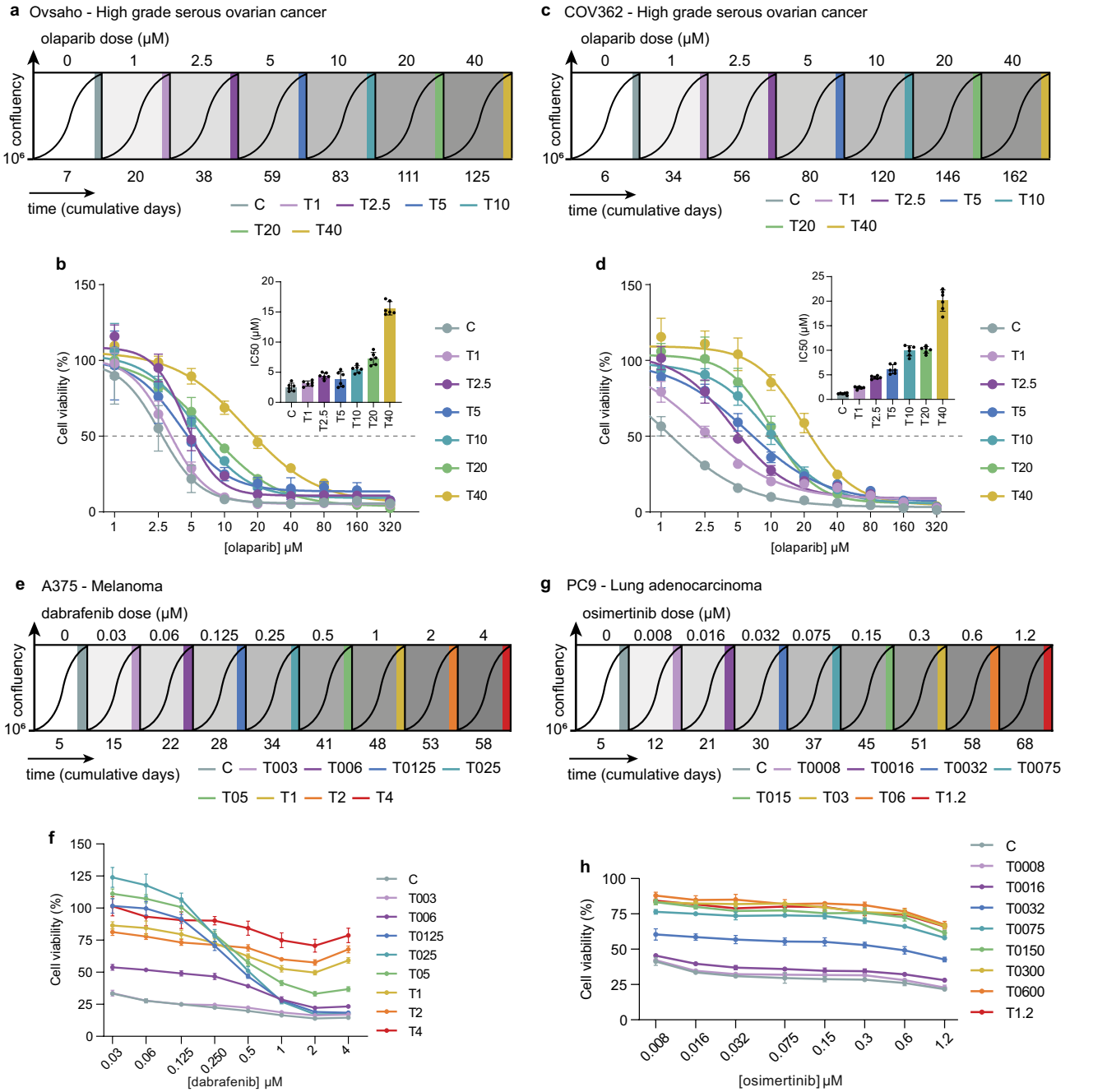
Additional information

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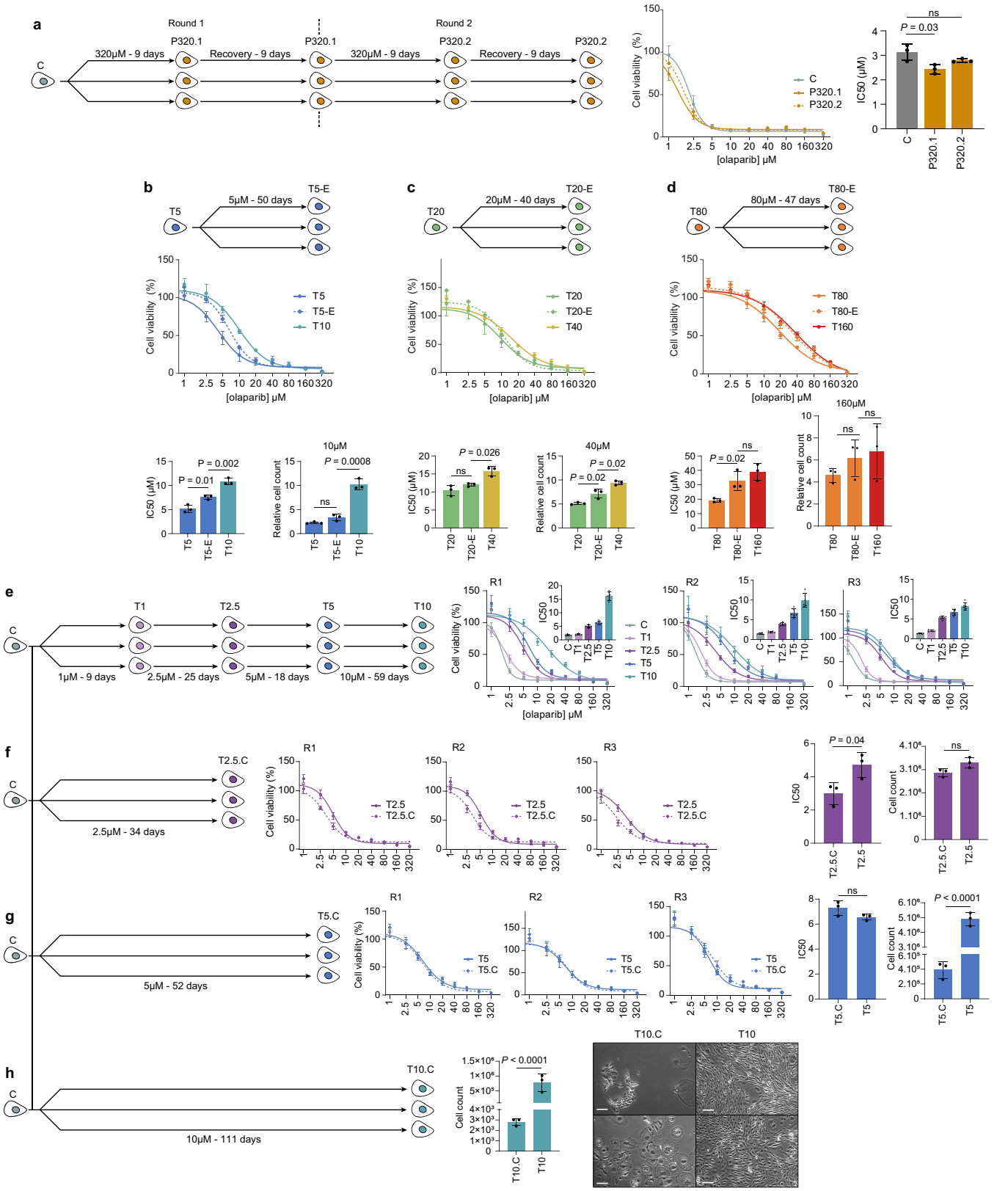
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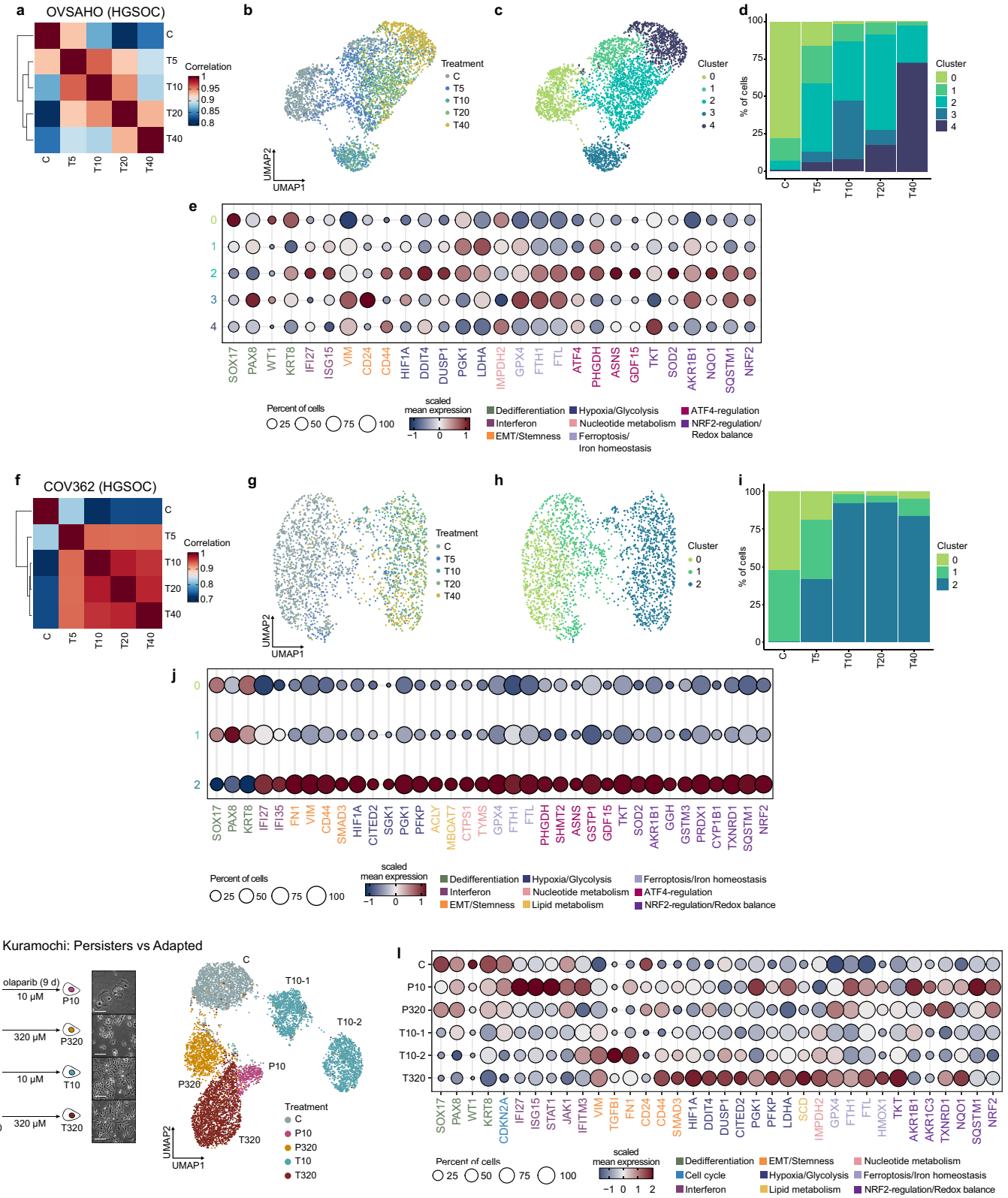


Extended Data Fig. 1 | Generation of drug-adapted populations in additional resistance models. a, Drug-adapted lines generated for Ovsaho, a BRCA-mutant ovarian cancer cell line, using dose-escalation from 1 to 40 μM of olaparib. **b**, Cell viability assays (9 days of treatment) for the Ovsaho adapted lines showing progressive increase in resistance. Bar plots show the average of 6 technical replicates with standard deviation (SD) error bars. **c**, Same as in **a** for COV362, also a BRCA-mutant ovarian cancer cell line. **d**, Same as in **b** for COV362. **e**, Drug-adapted lines generated for A375, a melanoma cell line, using dose-escalation from 30 nM to 4 μM of dabrafenib. **f**, Cell viability assays (4 days of treatment) for the A375 adapted lines. **g**, Drug-adapted lines generated for PC9, a lung adenocarcinoma cell line, using dose-escalation from 8 nM to 1.2 μM of osimertinib. **h**, Cell viability assays (5 days of treatment) for the PC9 adapted lines. Cell viability assays were performed using CellTiter Glo 2.0 and values are normalized by DMSO control. Data are shown as means and SDs of 6 technical replicates.



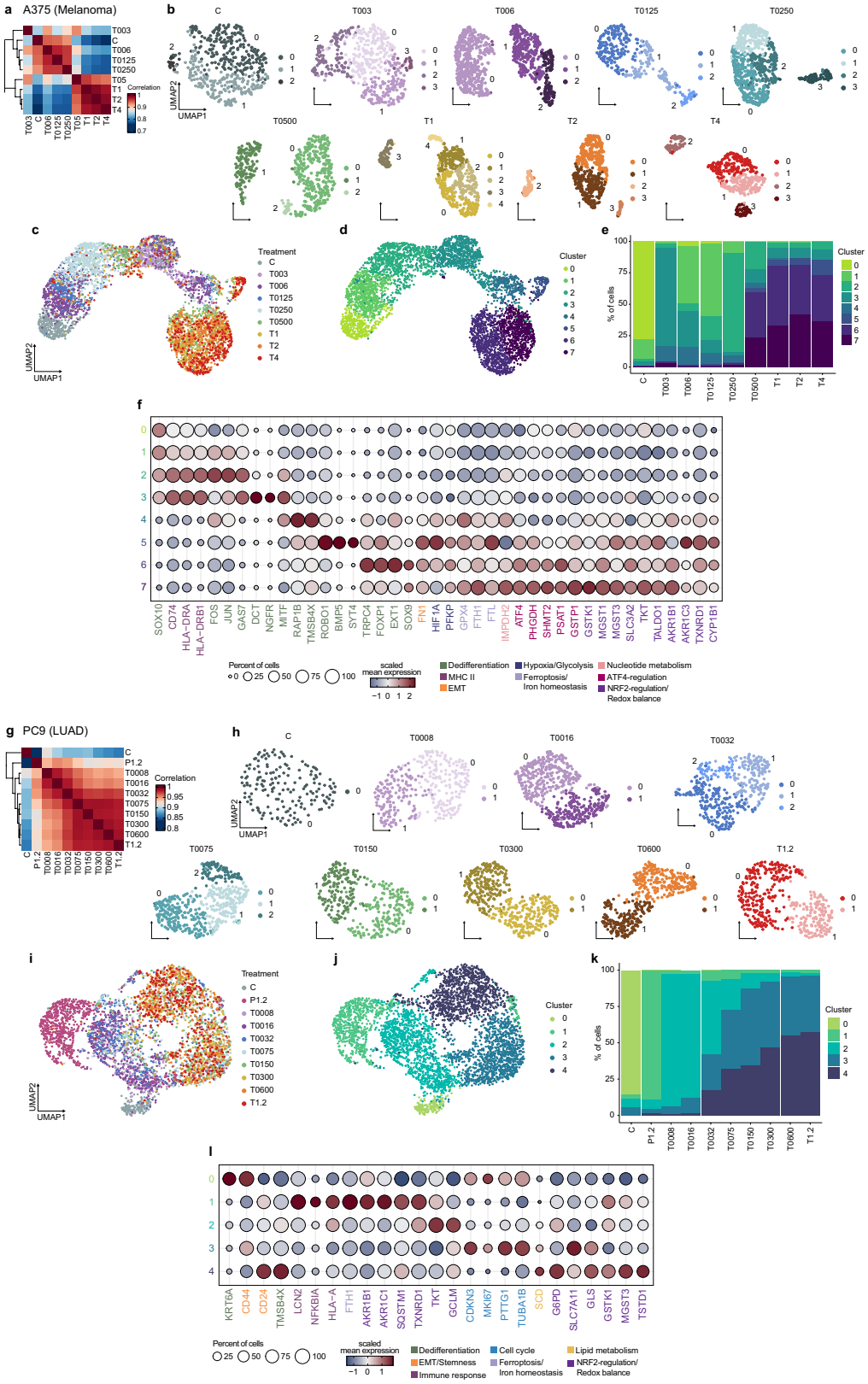
Extended Data Fig. 2 | See next page for caption.

Extended Data Fig. 2 | Drug-adaptation in acute versus gradual dose exposure and treatment duration. a, Generation of drug-tolerant persisters for Kuramochi cell line by treatment with 320 μM of olaparib for 9 days. The surviving cells were allowed to recover for 9 days without treatment. Two rounds of treatment-recovery experiments were performed in triplicates and the viability of the recovering cells was tested. Cell viability curves are shown as the average of three replicates (n = 6 technical replicates) and error bars as SD. The bar plot shows no significant IC_{50} increase for persisters relative to the parental population (C vs P320.1: $P = 0.03$; ns, not significant; two-tailed t-tests). **b**–**d**, Extended treatments at constant doses. Representative Kuramochi adapted lines (T5, T20 and T80) were maintained on olaparib treatment at the final concentrations to which they had adapted (5, 20 and 80 μM, respectively) for the number of days that they had taken to adapt to their subsequent populations at increased concentrations (T10, T40, T160, respectively) (see Fig. 1a), generating the extended T5-E, T10-E and T80-E populations (three independent replicates). Prolonged exposure at constant doses did not generate similarly resistant populations for the T5 and T20 extended treatments when compared to their dose-escalated counterparts (T10 and T40, respectively). The extended treatments did however lead to higher resistance relative to their parental populations. The cells from the T80 extended treatments were similarly resistant to T160, highlighting the role of treatment duration. Together, these observations indicate that both dose and time influence the fitness outcomes along the adaptive path. Cell viability was measured by CellTiter Glo 2.0 (each biological replicate with 6 technical replicates) and by cell counting (bar plots show the average of three biological replicates and error bars as SD) after 9 days of treatment at their respective escalated doses (10, 40 and 160 μM). IC_{50} (left bar plots) and relative cell counts (right bar plots) were compared with their parental and dose-escalated populations (significant *P*-values are shown; ns, not significant; two-tailed *t*-tests). Dose response curves for the original T5, T10, T20, T40, T80 and T160 are from Fig. 1b as performed in 3 independent experiments. **e**–**h**, To test whether dose escalation facilitates drug adaptation in our model, we compared the resistance phenotypes between cells that were dose-escalated from 1 to 10 μM with drug-naive cells directly exposed to the same doses for the same duration. Dose-escalated cells were able to proliferate and reach confluency, contrasting with drug-naive cells directly exposed to higher doses (5 and 10 μM). The continuous long-term 10 μM treatment led to very few viable cells, which prevented us from testing their fitness with viability assays (**h**). The populations that were dose-escalated up to 2.5 μM were significantly more resistant than their continuously treated counterparts. Plots indicate dose-response curves for three biological replicates (R1, R2, R3; 6 technical replicates each) and bar plots of average IC_{50} values (3 biological replicates with SD error bars). Bar plots (right) showing the average cell counts in **f**-**h** for the 3 biological replicates with SD error bars. Cell viability assays and cell counts (treated with 2.5, 5 and 10 μM of olaparib, respectively) after the treatment periods are shown and compared with the dose-escalated populations (*P*-values are indicated; ns, not significant; two-tailed *t*-tests). Microscopic photos show that T10 (dose-escalated) cells reached confluency while T10.C (continuous) did not (scale bar 50 μm).



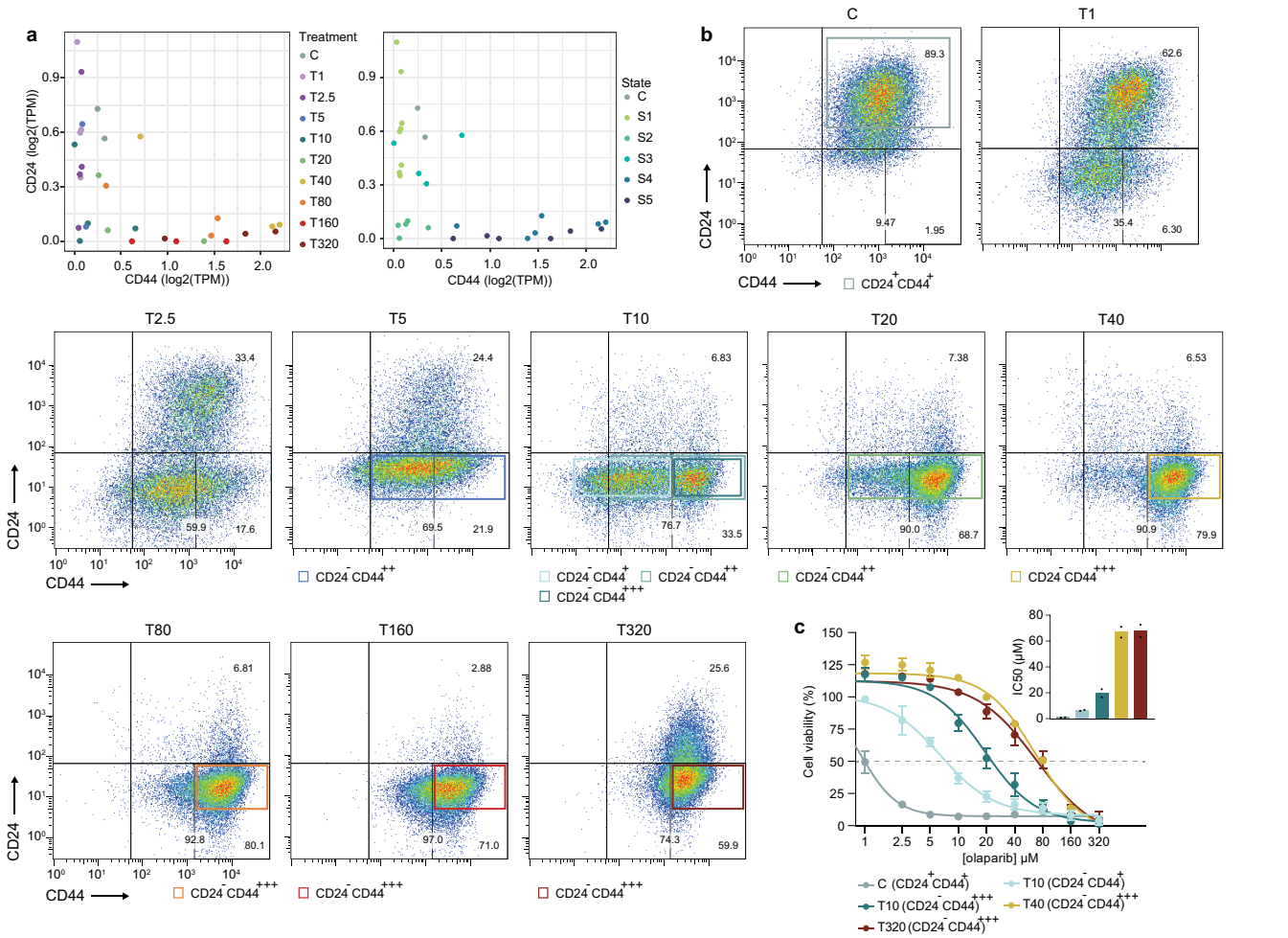
Extended Data Fig. 3 | See next page for caption.

Extended Data Fig. 3 | scRNA-Seq analysis for two additional ovarian cancer adapted cell lines. a, Spearman's correlations across the averaged transcriptomes of the Ovsaho olaparib-adapted lines. Untreated control (C) and T5 to T40 were collected for scRNA-Seq. **b-c**, UMAP of Ovsaho scRNA-seq data on the adapted lines colored by treatment (b) and by cluster as determined by Louvain clustering (c), highlighting the emergence of distinct cellular states. **d**, Bar plot representing the percentage of cells from each condition in each cluster, indicating cell state frequency changes across the adapted lines. **e**, Differentially expressed genes across Ovsaho clusters, highlighting the expression of shared genes with Kuramochi (Fig. 2a), COV362 (Extended Data Fig. 3j) and PDX resistance models (Fig. 4f). Color bar represents the scaled average expression for a particular cluster and circle sizes are the fraction of cells expressing the gene. Overall, genes highlight dedifferentiation and metabolic reprogramming along the resistance continuum. **f**, Same as in **a** for COV362. **g-h**, Same as in **b** and **c** for COV362, respectively. **i-j**, Same as in **d** and **e** for COV362, respectively. **k**, Drug-tolerant persisters generation by treatment of drug-naive cells with 10 and 320 μM of olaparib for 9 days. The adapted T10 and T320 lines were maintained at the same doses, respectively (left, scale bar 50 μm, representative of two biological replicates). UMAP of the scRNA-Seq data for the persister and adapted cell populations (right). Phenotypic features of persisters are shown in Supplementary Fig. 2. **l**, Differentially expressed genes across conditions (relative to C) highlighting shared and distinct marker gene expression among the persister and adapted lines.

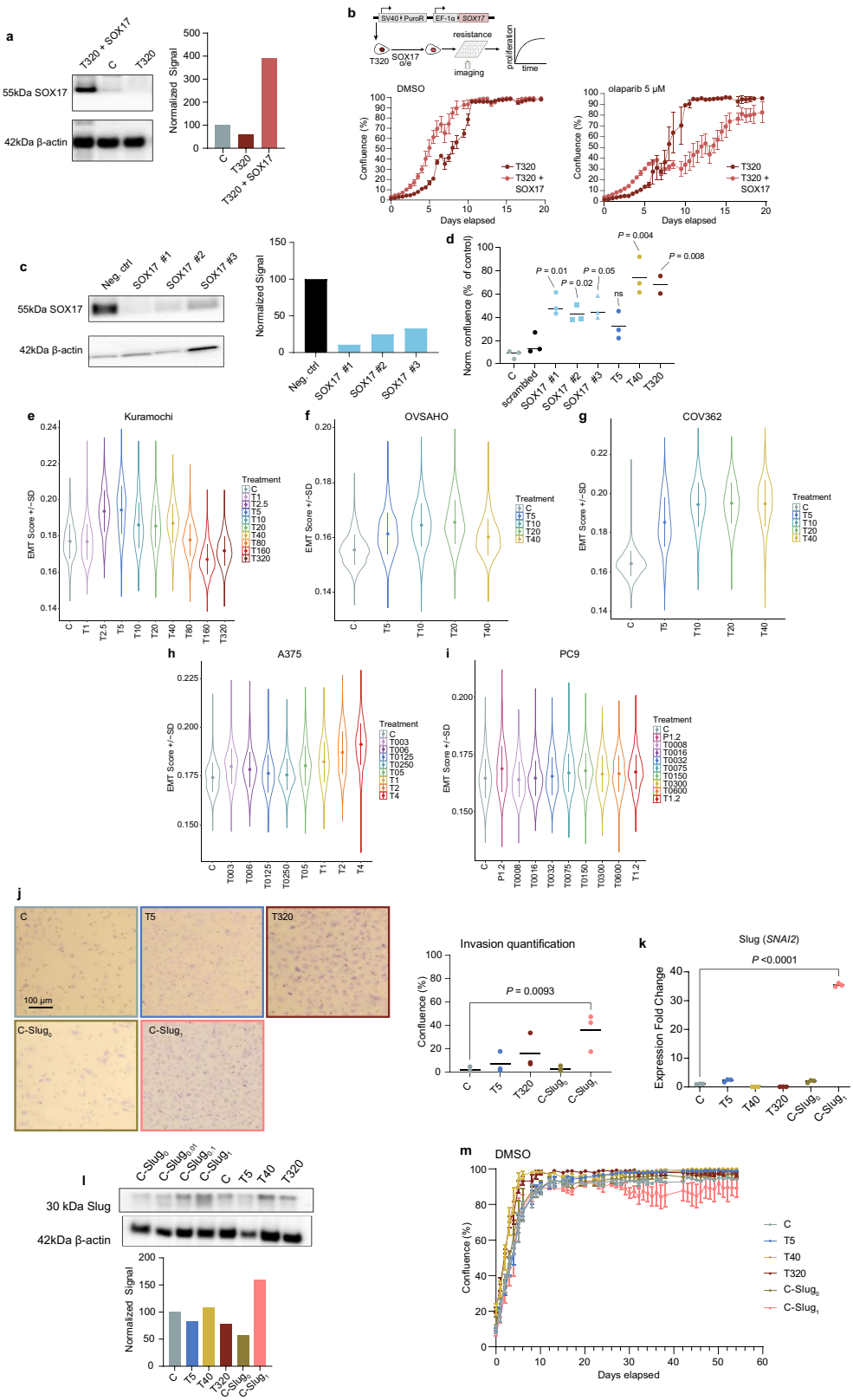


Extended Data Fig. 4 | See next page for caption.

Extended Data Fig. 4 | scRNA-Seq analysis for the melanoma and lung adenocarcinoma adapted lines. a, Spearman's correlations across the averaged transcriptomes of the A375 (melanoma) adapted lines that evolved resistance through dose-escalation from 30 nM to 4 μM of dabrafenib (Extended Data Fig. 1e,f). **b**, UMAP of scRNA-seq data on the individual lines. Colors and numbers indicate subpopulations determined by Louvain clustering. **c**-**d**, UMAP for all cells colored by treatment (c) and by cluster as determined by Louvain clustering (d), highlighting distinct transcriptional states across the adapted lines. **e**, Bar plot representing the percentage of cells from each condition in each cluster, indicating the emergence of cell states and frequency changes that suggest state selection along the resistance continuum. **f**, Differentially expressed genes across A375 clusters, highlighting the expression of lineage plasticity and metabolic reprogramming markers across the resistance continuum. Differentiation markers indicate multiple intermediate states, showing a progression from *SOX10*^{high} (clusters 0-3) to *SOX10*^{low} expression (clusters 4-7) and loss of MHC class II genes (*CD74*, *HLDA-DRA/B*) (Supplementary Table 4). Diverse lineage states arise along the resistance path, such as cells expressing both melanocytic (*DCT*, *MITF*) and neural-crest (*NGFR*) lineage markers (cluster 3); a population expressing highly *MITF*, *RAP1B* and *TMSB4X* (cluster 4); cells expressing neuronal markers (*ROBO1*, *BMP5*, *SYT4*) (cluster 5); and finally a mesenchymal-like invasive state (*SOX9*, *FN1*) (clusters 6-7, Supplementary Table 4). A gradual increase in the expression of ATF4 and NRF2 regulated genes toward the most resistant states (clusters 4 to 7), indicating extensive metabolic rewiring. Color bar represents the scaled average expression for a particular cluster and circle sizes are the fraction of cells expressing the gene. **g**-**h**, Same as in **a** and **b** for PC9 (lung adenocarcinoma), respectively. Adapted lines evolved resistance through dose-escalation from 8 nM to 1.2 μM of osimertinib (Extended Data Fig. 1g,h). A persister population (P1.2) was generated by treatment with 1.2 μM of osimertinib for 9 days. **i**-**l**, Same as in **c**-**f** for PC9, respectively, including a persister population (P1.2, **i**). PC9 cells transitioned to a *KRT6A*^{low} low expression and coexist in a combination of CD44^{high}CD24^{low} and CD44^{low}CD24^{high} states (**l**) (Supplementary Table 5).

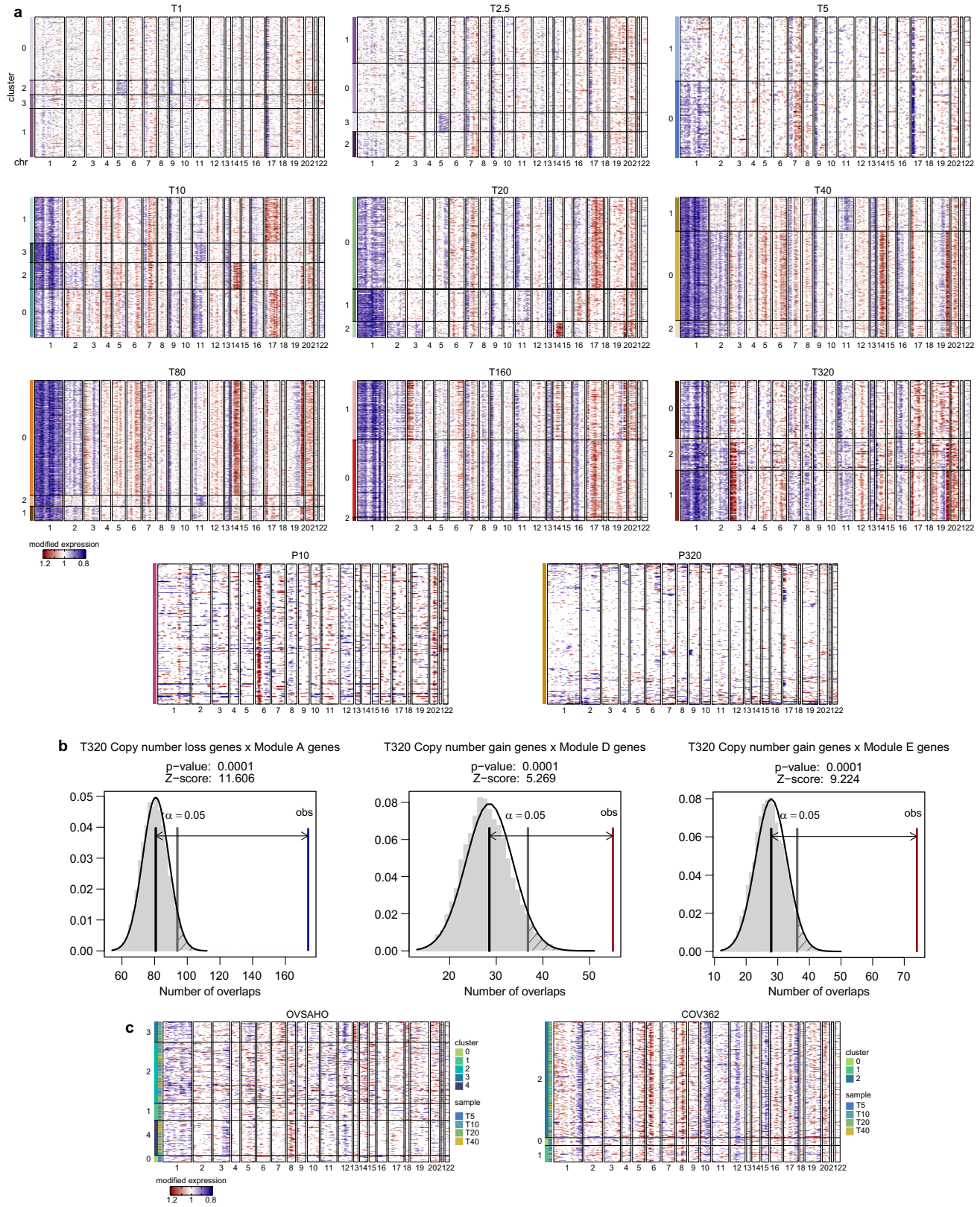


Extended Data Fig. 5 | Expression of CD24 and CD44 as a proxy for adapted states in Kuramochi cells. a, Scatter plot showing the average expression of CD24 and CD44 markers in each of the identified subpopulations colored by treatment condition (left) and state (right). **b**, Representative plots of the flow cytometry analysis of CD24 and CD44 staining across the Kuramochi adapted populations. Bottom left quadrants represent double negative staining for both markers. A transition from $CD24^{high}CD44^{low}$ (top right) to $CD24^{low}CD44^{high}$ (bottom right quadrants) is observed. Numbers indicate the percentage of cells in each quadrant. Colored squares indicate the subpopulations sorted for the cell viability assays shown in (c), ATAC-Seq (Fig. 3) and CRISPR-based screen (Fig. 5). **c**, Cell viability assays of sorted subpopulations indicating differential resistance within (T10) and between (T10, T40, T320) treatment conditions. Sorted subpopulations were labeled according to their CD24/CD44 profiles. Bar plots show the IC_{50} averages of two independent experiments (6 technical replicates each).



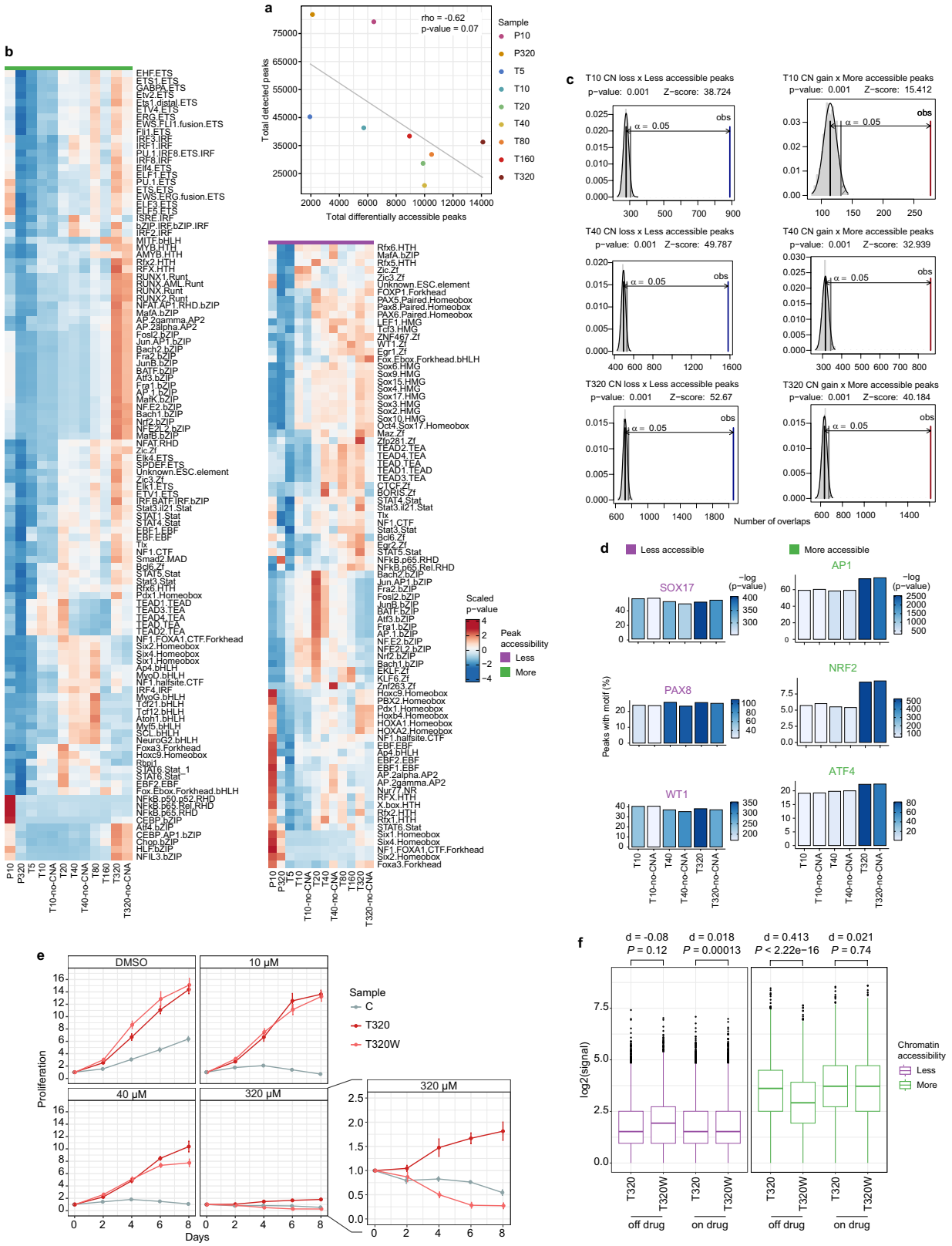
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Extended Data Fig. 6 | EMT signatures and perturbation of lineage plasticity in drug resistance. a, Western blot of the induced SOX17 in the T320 line compared to the parental control (C) cells (three technical replicates). Bar plot shows the quantification signal. Full western blot shown in Supplementary Fig. 6. **b**, Schematic of the constitutive SOX17 overexpression construct (top). Quantification by live cell imaging of the percent area covered by cells over a 20 day time period of T320 parental and T320 with *SOX17* overexpression on DMSO (left) and on olaparib (5 μM) treatment (right). Average and SD of three biological replicates (three technical replicates each) shown. **c**, Western blot of SOX17 shRNA knockdown in parental control (C) cells compared to non-targeting scrambled shRNA (three technical replicates). Bar plot shows the quantification signal. Full western blot shown in Supplementary Fig. 7. **d**, Quantification of SOX17 shRNA knockdown in parental control (C) cells and adapted T5, T40 and T320 by live cell imaging (three biological replicates, three technical replicates each) of the percent area covered by cells normalized by DMSO controls on day five of treatment of 2.5 μM olaparib. Standard two-sided t-test was performed by comparing with the negative control cells (Neg. Ctrl) (SOX17 #1: $P = 0.01$; SOX17 #2: $P = 0.02$; SOX17 #3: $P = 0.05$; T40: $P = 0.004$; T320: $P = 0.008$; ns: not significant). **e-i**, EMT signature scores were calculated across treatment conditions for each resistance model. The set of genes annotated as "Hallmark epithelial-mesenchymal transition" from the MSigDB was used to calculate the scores with the UCell algorithm. Violin plots represent the score distributions with the average plus and minus SD. Numbers of cells are: **(e)** C (N = 453), T1 (N = 1223), T2.5 (N = 760), T5 (N = 398), T5 (N = 398), T10 (N = 1030), T20 (N = 640), T40 (N = 918), T80 (N = 910), T160 (N = 1348), T320 (N = 568); **(f)** C (N = 840), T5 (N = 713), T10 (N = 704), T20 (N = 731), T40 (N = 650); **(g)** C (N = 1148), T5 (N = 180), T10 (N = 278), T20 (N = 447), T40 (N = 272); **(h)** C (N = 442), T003 (N = 478), T006 (N = 906), T0125 (N = 306), T0250 (N = 733), T05 (N = 493), T1 (N = 605), T2 (N = 641), T4 (N = 455); **(i)** C (N = 169), P1.2 (N = 673), T0008 (N = 329), T0016 (N = 528), T0032 (N = 414), T0075 (N = 505), T0150 (N = 329), T0300 (N = 442), T0600 (N = 424), T1.2 (N = 395). **j**, Representative images (biological duplicates) of stained membranes from a transwell cell invasion assay. Colors represent the untreated cells (C), adapted lines T5 and T320 and the Slug-lines on different doxycycline (dox) concentrations (C-$Slug_0$ as negative and C-$Slug_1$ on 1 μg/ml of dox). Quantification of transwell cell invasion assay data of three images per cell line (right). One way ANOVA comparing Slug-induced (*SNAI2*) lines to the drug-naive (C) cell line. **k**, Quantification of induced *SNAI2* gene expression from qRT-PCR (two independent runs in triplicates). One way ANOVA comparing Slug-induced lines to the drug-naive (C) cell line. **l**, Representative western blot (three technical replicates) of the induced Slug lines with different doxycycline concentrations ranging from 0.01 μg/ml to 1 μg/ml. We observed that C-$Slug_1$ produced a relevant phenotype, which was used for further experiments. Bar plot shows the quantification signal. Full western blot shown in Supplementary Fig. 8. **m**, Quantification by live cell imaging of DMSO controls For Slug-induced lines. Data represents percent area covered by cells over a 54 day time period. Average and SD of three biological replicates (three technical replicates each) are shown.



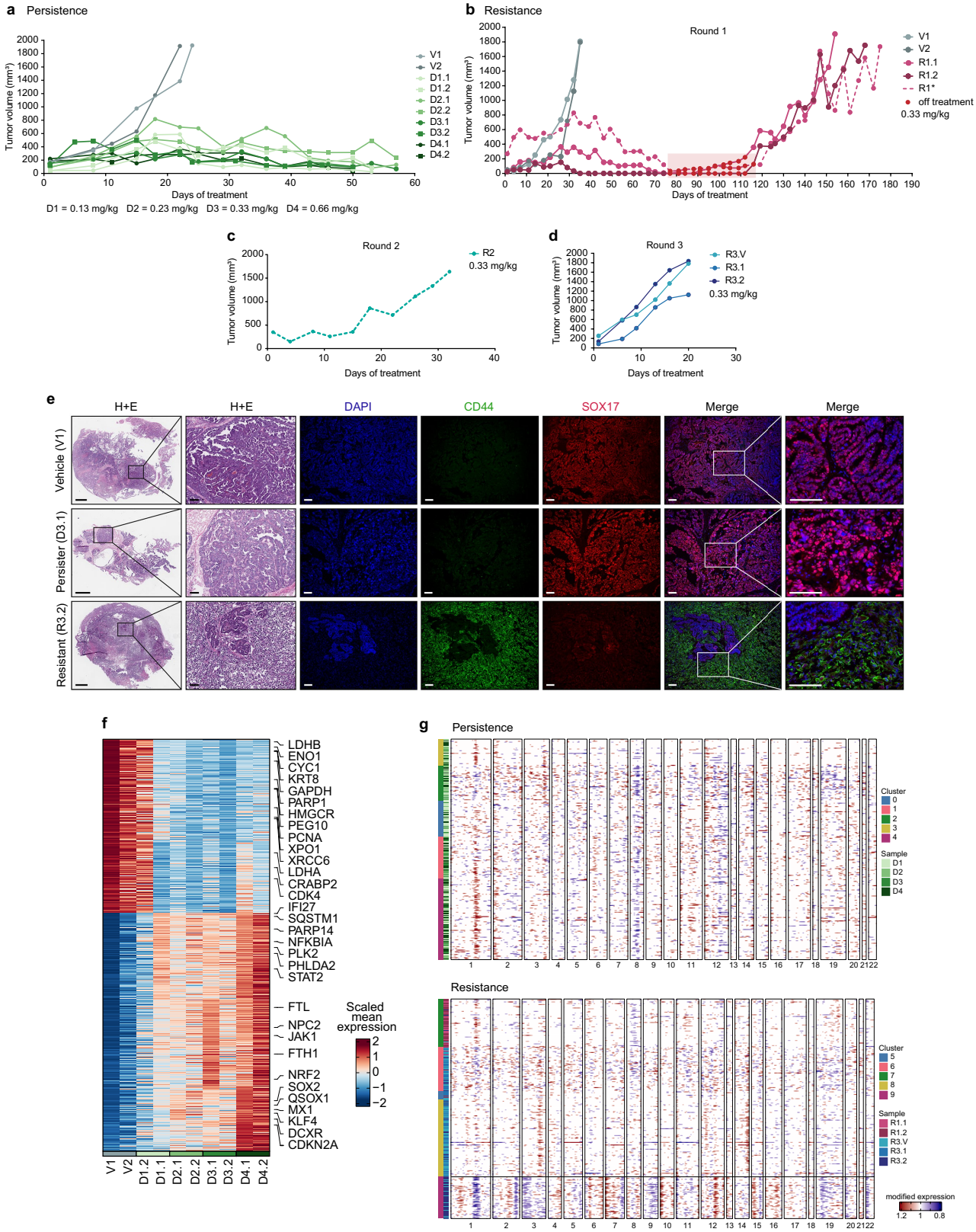
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Extended Data Fig. 7 | Copy number inference from scRNA-seq data for the adapted and persister lines. a, Heat maps depicting scRNA-Seq inferred copy number alterations (CNAs) for Kuramochi adapted and persister lines relative to untreated control (C). Gene expression signal is scaled along the chromosomes as calculated by inferCNV. Red represents genomic regions with inferred copy number gain and blue represents regions with inferred copy number loss. Colors and labels on the left represent the subpopulations identified by transcriptome clustering (Fig. 1d) showing correspondence between inferred CNAs and transcriptional states. **b**, Enrichment of genes within CNA regions identified by whole-exome sequencing overlapping with the genes present in the transcriptional modules detected for the adapted lines (Fig. 2a) in respect to random distributions (*P*-values determined by 10,000 permutations, two-sided permutation tests). Blue and red lines indicate the observed overlap of copy number loss and gain, respectively, and the gray lines indicate the significance threshold in an expected random distribution. **c**, Inferred CNAs defined for the adapted Ovsaho and COV362 lines relative to their untreated controls. Color annotations represent sample and transcriptional clusters identified in Extended Data Fig. 3c,h.



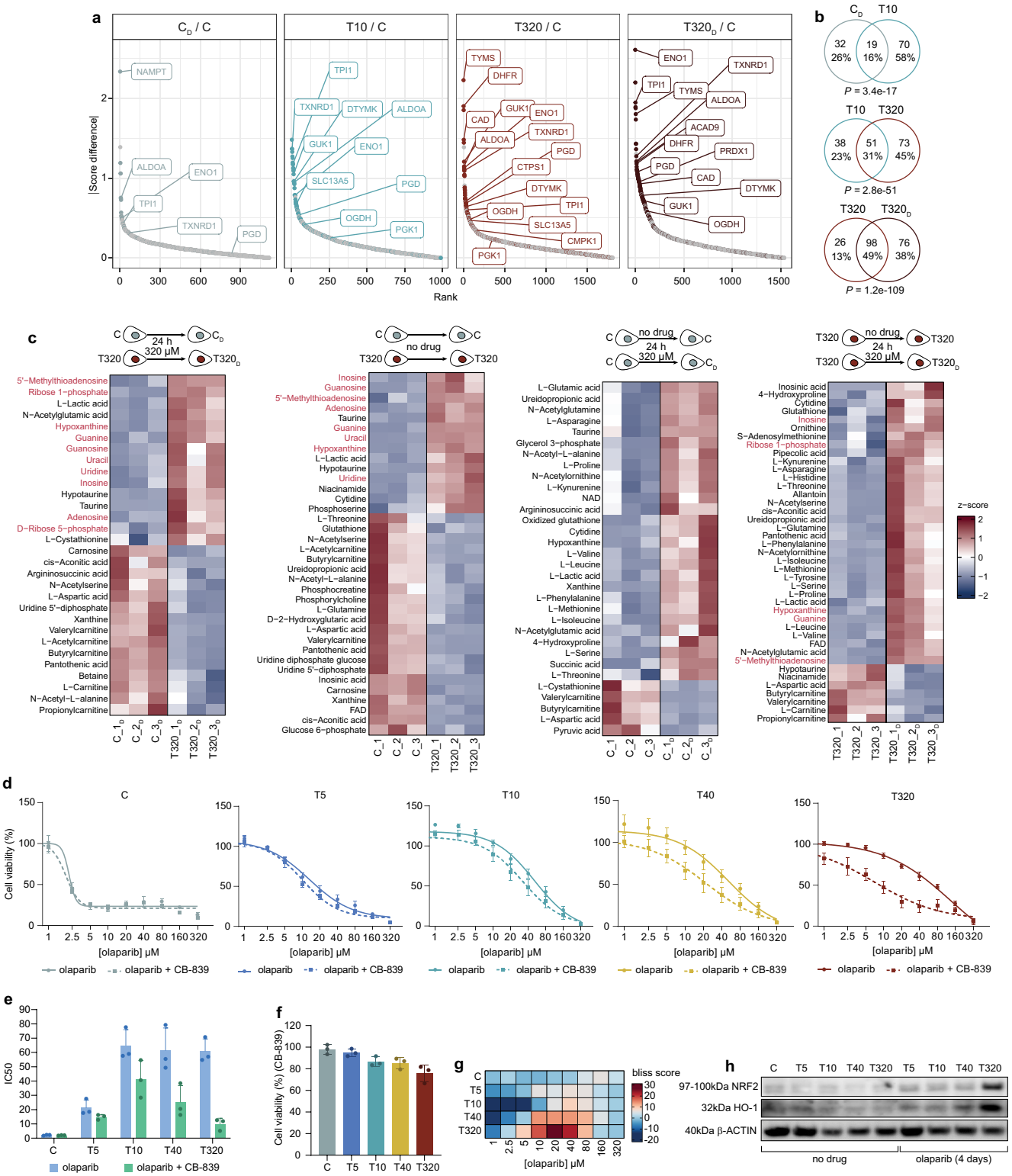
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Extended Data Fig. 8 | Chromatin accessibility and stability of drug resistance. **a**, Correlation between the number of differentially accessible peaks and total number of detected peaks per sample. The lack of a positive correlation indicates that the number of differentially accessible peaks are not explained by peak detection bias (Spearman's rho = −0.62, *P* = 0.07, two-sided Spearman Rank Correlation test). **b**, Heat maps showing the hierarchical clustering (rows) of the enriched TF motifs in the differentially more (green) and less (purple) accessible peaks (relative to C). Color bar represents the scaled p-values for TF motif enrichment (only transcription factors with *P*-value < 1e-5 were included, one-sided hypergeometric test, Benjamini-Hochberg corrected). The analysis controlled by peaks not overlapping CNAs ("no-CNA") reveals the same patterns when compared to all peaks. **c**, Enrichment of differentially accessible ATAC-Seq peaks overlapping CNA regions (relative to the untreated control C) identified by whole-exome sequencing for representative adaptive lines (T10, T40 and T320) (*P*-values determined by 1,000 permutations, two-sided permutation tests). Blue and red lines indicate the observed overlap of copy number loss and gain, respectively, and the gray lines indicate the significance threshold from the expected random distribution. The proportion of peaks not overlapping CNAs ranged from 70% for less, and 80–90% for more accessible peaks, highlighting widespread global chromatin changes. **d**, Transcription factor (TF) motif enrichment analysis controlling for peaks not overlapping with CNAs, showing similar frequency of peaks containing TFs motifs involved with regulation of stress response (AP1, NRF2 and ATF4) and ovarian cancer TFs (SOX17, PAX8 and WT1). Bar plots represent the percentage of differentially accessible peaks for each condition (relative to C). Color bars represent the motif enrichment *P*-values (one-sided hypergeometric test, Benjamini-Hochberg corrected). **e**, Proliferation of C, T320 and T320W (which is the T320 population after drug removal for 120 days) on DMSO, 10, 40 and 320 μM of olaparib for 8 days. The T320W exhibits partial reversibility in resistance, especially at 320 μM. Proliferation was measured every 2 days using Cell TiterGlo 2.0 assay and values are normalized from day 0 (24 h after cell seeding). Data represents the average (dots) of 6 technical replicates and error bars indicate the SD. **f**, Box plots of the signal for less (purple) (N = 6771) and more (green) (N = 7332) accessible peaks shown in Fig. 3f. This shows that the signal across the less accessible regions (relative to C, Fig. 3d) both with and without treatment are overall stable when comparing T320 and T320W. For the more accessible regions, the signal decreases in T320W without treatment and recovers its accessibility during drug treatment. *P*-values were determined from two-sided Wilcoxon tests and the effect sizes are Cohen's d. Boxes represent the interquartile range (IQR) and centre lines represent the median. Whiskers extend to the smallest and largest values within 1.5 times the IQR from the lower and upper quartiles, respectively. Data points beyond the whiskers are outliers.



Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | Persistence and resistance model of high-grade serous ovarian cancer patient-derived xenografts. a, Tumor growth curves for PDXs treated with four doses of talazoparib (shades of green) and vehicle controls (gray) in the persistence experiment. Each condition was collected in duplicates for scRNA-Seq. Two replicates of each condition were generated (indicated by ".1" and ".2"). **b-d**, Tumor growth curves for PDXs in first (b), second (c) and third (d) rounds of talazoparib treatment in the resistance experiment. A drug holiday in the first round is indicated in red. Two replicates for each condition were collected for scRNA-Seq: vehicles (gray), round 1 (solid pink) and round 3 (blue, with an additional vehicle treated R3.V). A replicate tumor from round 1 (dashed pink) was seeded in a new cohort to generate the round 2 treatment. **e**, Hematoxylin and eosin (H + E) staining of the entire section of V1, D3.1 (persistence) and R3.2 (resistance) (column 1) and an enlargement (column 2). Histology was annotated by a gynecologic pathologist. The V1 tumor has the classic appearance of serous carcinoma with nests of neoplastic cells with "slit-like spaces" separating strands of tumor cells. The D3.1 tumor still is recognizable as serous carcinoma while more poorly differentiated than V1. The R3.2 tumor has a biphasic pattern with some areas that retain a recognizable papillary pattern, while it is dominated by poorly differentiated/undifferentiated tumor structures with discohesive tumor cells. DAPI staining (blue), CD44 (green) and SOX17 (red) as single channels (columns 3-5), merged (column 6) and an enlargement (column 7). Consistent with scRNA-Seq data, the well-differentiated regions in V1 highly express SOX17 and lack expression of CD44, which is also observed in D3.1. Regions of similar histology in R3.2 have weaker expression of SOX17 and remain to lack expression of CD44 (columns 4-5). In contrast, the poorly differentiated regions in R3.2 strongly express CD44 and weakly express SOX17 (columns 6-7), thus indicating heterogeneous populations in different resistant states. Scale bars represent 100 μm except in column 1 (1000 μm). **f**, Heat map of the scaled gene average expression across samples of the dose-response experiment (persistence), highlighting differentially expressed genes that show a dose-dependent trend. **g**, Inferred CNAs defined for the persistence and resistance PDX experiments relative to their respective vehicle treated tumors. Color annotations represent sample and transcriptional clusters identified in Fig. 4b,d.



Extended Data Fig. 10 | See next page for caption.

Extended Data Fig. 10 | Metabolic vulnerabilities along the resistance continuum. a, Rank plots showing the score differences for C_D, T10, T320 and $T320_D$. Higher absolute score differences indicate increased gene depletion relative to C. Highlighted genes are significantly depleted (FDR < 0.05, two-sided likelihood ratio test (LTR)) at the endpoint. Only genes with negative score differences are shown, indicating gene depletion. **b**, Overlap between the significantly depleted genes with high absolute score differences (>1 SD) relative to C. *P*-values were determined by one-sided hypergeometric tests. **c**, Heat maps of the differentially abundant metabolites ($P < 0.05$, two-sided t-tests) comparing olaparib sensitive (C) and resistant (T320) populations. Conditions represent cells on olaparib treatment (320 μM for 24 h) or cells without treatment. The experiment was performed in triplicates for each condition. Color scale represents scaled normalized metabolite abundance. Metabolites involved in nucleotide biosynthesis are highlighted in red. **d**, Dose-response curves of olaparib treatment or in combination with CB-839 after 6 days for representative populations within the resistance continuum (C, T5, T10, T40 and T320). Data points are normalized relative to vehicle-treated controls (for each respective line) and represent the average of 3 independent experiments (6 technical replicates per experiment) and SD. **e**, Bar plots showing the IC_{50} determined from dose-response curves **(d)** after 6 days of treatment with olaparib, or the combination of olaparib with CB-839. Dots represent three independent experiments (average of 6 technical replicates). Cells were sorted based on $CD24^{low}/CD44^{high}$ profiles (as in Extended Data Fig. 5b,c). **f**, Bar plot showing the average cell viability of cells treated with CB-839 (50 nM) normalized to vehicle controls (3 independent experiments, 6 technical replicates each). **g**, Heat map of the bliss synergy scores[41] of the olaparib and CB-839 treatment combination. **h**, Representative western blot (biological duplicates) showing the expression of NRF2 and its target HO1 (*HMOX1*) with and without olaparib treatment. Full western blot is shown in Supplementary Fig. 10.

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n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Data collection was performed using inDrop (pipeline v0.1.0), 10X Genomics Chromium kit v3.1 (pipeline v.3.1 and v.7.0.0).
Data analysis	Analysis of single-cell gene expression was performed in R 3.4.1 and 4.0.3 with Seurat versions 2.3.2 and 4.0.5. The following additional R packages were used: regioneR, fgsea, Mfuzz, GeneOverlap, UCell, DESeq2, inferCNV v1.4.0. Analysis of cell viability was performed using GraphPad Prism 9. Flow cytometry data was analyzed using BD Facs Diva v.8.0.2 and FlowJo v.10.8.1. ATAC-Seq data analysis was performed using HOMER 4.1, bowtie2 v2.3.4.1, SAMTools 1.9, IDR 2.0.3, Piccard Tools 2.18.2, trimgalore 0.5.0, deepTools v3.1. Whole exome sequencing data analysis was performed using Trimmomatic 0.36, BWA-MEM, bcl2fastq, Sambamba 0.6.8, Seq-N-Slide v21.1, Control-FREEC, ANNOVAR (170716) and GATK 3.8.1. CRISPR-Screen data analysis was performed using MAGeCK 0.5.8 (MAGeCK-count, MAGeCK-MLE). Relevant code is provided at: https://github.com/yanailab/drug-resistance

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Raw and processed data from this manuscript are publicly available in the GEO database under accession number GSE247691. Metabolomics data is publicly available in the Metabolomics Workbench (ST003102, http://dx.doi.org/10.21228/M8WM7T).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	n/a
Reporting on race, ethnicity, or other socially relevant groupings	n/a
Population characteristics	n/a
Recruitment	n/a
Ethics oversight	n/a

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	In vivo PDX experiments were performed with n = 2 biological replicates per condition. This sample size was selected based on ethical considerations and practical limitations associated with animal studies and reproducible effects were observed in downstream analyses. CRISPR screens were conducted on each biological sample, with reproducibility observed across different biological samples. Western blots for protein expression were performed in at least n = 2 independent experiments with reproducible patterns. Cell line treatment combinations were performed in 3 independent experiments. Dose-escalation experiments were performed in 5 different samples (cell lines) with at least 5 time points. Most cell viability and proliferation experiments were performed with n = 3 biological replicates and 6 technical replicates.
Data exclusions	For single-cell analyses: cells were excluded based on QC metrics as described in the methods section.
Replication	Most experiments were repeated at least twice (with n = 3) showing reproducible patterns. Mouse experiments included at least 2 replicates of each condition, showing reproducible patterns with or without talazoparib treatments. Gene expression patterns were used to assess reproducible trends across different models of in vivo and in vitro experiments.
Randomization	Mice were randomly assigned to experimental conditions before vehicle and drug treatments. For cell line based experiments, cells were seeded and wells randomly assigned.
Blinding	To analyze the data knowing the conditions for sample processing and analyses were necessary for results interpretation. Data collection was done blindly.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

TotalSeq A anti-human (anti-CD298 and anti-B2M, Biolegend), hashtag oligos 1:10 for scRNA-seq 10X cell hashing. Cat. Nos: 394601, 394603, 394605, 394607, 394609, 394611, 394613, 394615, 394617, 394619
anti-human CD44 (Cat No. 559942) and CD24 (Cat No. 555428) (BD Biosciences)
Annexin V PE (Apoptosis detection kit I Cat No. 559763, BD Biosciences). Goat anti-Rabbit immunoglobulin G (IgG) (H + L) horseradish peroxidase (HRP) (#31460), Goat anti-Mouse IgG (H + L) HRP (#31430) and rabbit anti-mouse (#A-11059) were used as secondary antibodies (Invitrogen, Thermo Fisher Scientific). Anti-beta Actin antibody (Santa Cruz, #sc-47778 or mAbcam 8226) was used as a loading control for all of the following staining, the NRF2 (Cell Signaling Technology (CST), #12721), ATF4 (CST, #11815), HO-1 (HMOX1, #MA1-112), Slug (CST, #95855), CD44 (R&D, MAB7045-SP) and SOX17 (CST, #8177, R&D AF1924-SP).

Validation

All these antibodies were validated by the manufacturer for western blot, flow cytometry and IHC/IF applications with human reactivity and used standardly in several previous publications (Stoeckius et al., Genome Biol. 2018; Chen et al., Nat. Metab. 2022; Chaves-Moreira et al. Sci Signal 2022; Lombard et al. Sci Adv 2020; Solier et al. Nature 2023; Reymond Sutandy, et. al Nature 2023; Chamseddine et al., Cell Rep 2020, Jung et al. Stem Cell Reports 2015; Patel et al. Onc. Lett. 2021; Valcourt et al. Cell Rep. 2021).

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)

Original source for Kuramochi and Ovsaho cell lines are commercially available from JCRB Cell Bank (Japansese Collection of Research Bioresources Cell Bank: JCRB0098 and JCRB1046). COV362 is available from Millipore Sigma (07071910). The A375 cell line is commercially available from ATCC (CRL-1619) and the PC9 cell line available from Millipore Sigma (90071810).

Authentication

Authentication confirmed by genotyping using short tandem repeat fingerprinting

Mycoplasma contamination

Cell lines were tested negative for Mycoplasma

Commonly misidentified lines
(See ICLAC register)

No common misidentified lines were used.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals

Immunodeficient six-week-old female NOD SCID gamma (NSG) (NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ) were purchased from Jackson Laboratory. Animals were housed in an SPF-grade animal facility at 25 ± 2°C with a humidity of 40%-70%. Besides, 12-hour/12-hour was alternated between day and night, and these mice were given an unrestricted diet. Before the experiment, these mice were adaptively reared for more than three days.

Wild animals

No wild animals were used in the study

Reporting on sex

Six-week old female mice were used in the study.

Field-collected samples

No field-collected samples were used in the study.

Ethics oversight

The study protocol was approved by the NYU Langone IACUC board.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	n/a
Novel plant genotypes	n/a
Authentication	n/a

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	- Annexin V/DAPI staining: To quantify apoptosis and cell viability of DTPs treated with different doses, we generated persister cells with either 10 or 320 µM of olaparib as previously described. The adherent cells were harvested (0.25% Trypsin/EDTA for 5 min at 37°C), stained with trypan blue and counted manually using hemocytometer or on a Countess II cell counter. Cells were washed twice in cold PBS and stained with DAPI (ThermoFisher) and Annexin V using the Annexin V Apoptosis Detection Kit (BD Biosciences) according to the manufacturer's instructions. - CD24/CD44 sorting: The untreated and resistant Kuramochi populations were harvested as above and stained with DAPI (Cat No. D1306, ThermoFisher), anti-CD24-PE (Cat No. 555428) and anti-CD44-APC (Cat No. 559942) (BD Biosciences).
Instrument	Flow cytometry analyses were performed on a LSR II Flow Cytometer (BD Biosciences), and Cell sorting on a Sony SY3200 Highly Automated Parallel cell sorter.
Software	Analysis performed on FlowJo v10.8.1 and BD Facs Diva 8.0.2
Cell population abundance	- Annexin V/DAPI staining: Double negative for Annexin PE / DAPI: 96.7% for untreated control cells, 25% for 10uM and 38% for 320uM treated cells (Kuramochi); Double negative for Annexin PE / DAPI: 96.7% for untreated control cells, 25% for 10uM and 38% for 320uM treated cells (COV362); Double negative for Annexin PE / DAPI: 96.7% for untreated control cells, 25% for 10uM and 38% for 320uM treated cells (Ovsaho). - CD44-APC, CD24-PE sorting: Untreated control (C) CD24+/CD44+: 80%; T5 CD24-/CD44+: 66.5%; T10 CD24-/CD44++: 72%; T20 CD24-/CD44++: 68%; T40 CD24-/CD44+++: 80%; T80 CD24-/CD44+++: 80%; T160 CD24-/CD44+++: 71%; T320 CD24-/CD44+++: 60%
Gating strategy	Live cells for annexin V/Dapi staining were gated based on FSC-A (18K-280K)/SSC-A (6.2K-250K); singlets were gated based on FSC-W (65K-144K)/FSC-A (14K-275K); DAPI (5.6K-84K MFI) and Annexin V PE (4.1K-249K MFI). CD24/CD44 staining: Live cells were gated based on FSC-A (3K-66K)/SSC-A (2K-36K); Single cells were gated based on FSC-H (2.4K-61K)/FSC-A (2.1K-63K); CD44-APC positive (54-67K), CD24-PE positive (65-62K); Untreated Ctrl CD44-PE (54-30K)/CD24-PE (173-40K); T5-T20 CD44-APC+ (54-62K) /CD24-PE- (0-65); T40-T320 CD44-APC+++ (1.5K-67K)/CD24-PE- (0-65) units in MFI.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.